

Supervised Bayesian Matrix Factorization

Joint Gene Expression Programs & Survival Modeling in PDAC

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Outline

Motivation · The PDAC challenge & why supervised factorisation matters

Methods · SBMF model · Dual-source objective · EBNM adaptive priors · CAVI inference

Results · Synthetic validation · Seven PDAC cohorts · Prior $\times$ K comparison · Deep dives

Future Directions & Summary

Overview of today's talk. About 15 minutes of content, then open for discussion. I'll start with why we need supervised approaches for PDAC, walk through the Bayesian model and inference, then present results on 7 independent cohorts.

The PDAC Problem

Pancreatic Ductal Adenocarcinoma

- **Five-year survival $< 12\%$** — one of the most lethal solid tumors
- Most patients diagnosed at advanced stage with limited treatment options
- **No widely adopted molecular stratification** at diagnosis
- Tumors are a complex mixture of malignant, stromal, and immune cells — gene expression signals are deeply interwoven
- **Goal:** identify *gene expression programs* (GEPs) whose activation levels predict patient survival

The Clinical Need

Reliable prognostic signatures could:

- Stratify patients for targeted therapy
- Define subgroups for clinical trial enrollment
- Distinguish molecular features that *describe* tumor heterogeneity from those that *predict* disease course

PDAC is the most lethal major solid tumor. Unlike breast cancer where molecular subtypes dictate treatment, PDAC patients largely receive the same regimen. If we can identify gene programs that distinguish prognostic subgroups, we can stratify patients for targeted therapy and clinical trials. The complexity is that bulk expression mixes signals from tumor, stroma, and immune cells.

Why Supervised > Unsupervised

The PCA-Then-Test Gap

Standard approach:

1. Apply PCA or NMF to expression data
2. Find axes of greatest expression variance
3. Test for survival association *post hoc*

The problem:

- Expression-dominant axes may reflect batch effects or housekeeping genes
- Programs with **modest expression variance** but **strong survival signal** get missed
- Post-hoc testing is statistically conservative

Supervised approach:

1. Integrate survival into the factorisation objective *directly*
2. Factors are simultaneously constrained to explain expression **and** predict survival
3. If $\hat{\beta}_k = 0$, factor k explains expression only — the model naturally separates prognostic from non-prognostic structure

💡 Key Insight

A factor explaining 15% of variance might be entirely batch effects. A factor explaining 3% might perfectly separate survivors from non-survivors. PCA prioritises the first and ignores the second.

PCA finds axes of greatest expression variance, not greatest clinical relevance. A survival-supervised approach forces factors to be useful for prediction. If a factor carries expression signal but no survival information, the model shrinks its beta to zero — we get the separation for free.

The Gap & Our Approach

What Existing Supervised MF Approaches Are Missing

- **Data-adaptive shrinkage priors** — existing methods use fixed penalties; we let the prior learn from the data
- **Automatic K selection** — no need to grid-search the number of factors
- **Uncertainty quantification** — posterior inference gives us credible intervals, not just point estimates

i SBMF vs. Penalised Supervised Approaches

Penalised approaches (e.g., deSurv from our lab) optimise a weighted objective $(1 - \alpha)\mathcal{L}_{\text{NMF}} + \alpha\mathcal{L}_{\text{Cox}}$, requiring a tuning parameter α to balance the two terms. **SBMF** instead uses full Bayesian inference:

- The **ELBO** naturally balances genomics and survival — no α needed
- **EBNM** priors adapt their shrinkage to the data
- Posterior second moments ($E[L^2] \neq \bar{L}^2$) prevent overfitting to uncertain loadings

SBMF: Learn gene expression programs and survival associations simultaneously, with the prior learning from the data.

This is the only slide where I reference deSurv — Amber’s work from our lab. The key distinction is that SBMF is fully Bayesian: we use the ELBO which naturally balances both objectives without a tuning alpha, and EBNM priors that adapt their shrinkage to the data. The posterior second moments are critical for preventing overfitting when loadings are uncertain.

The SBMF Model

Joint Genomics + Survival Factorisation

$$Y = LF^\top + E, \quad E_{ij} \sim N(0, \tau_j^{-1})$$

$$h_i(t) = h_0(t) \exp(L_i \cdot \beta)$$

i Baseline Hazard

$h_0(t)$ is **non-parametric** — it cancels exactly in the Cox partial likelihood. No shape assumption (Weibull, exponential) is made; the model only requires the proportional hazards structure.

Quantity	Dimension	Meaning
Y	$n \times p$	Gene expression matrix
L	$n \times K$	Patient activation scores
F	$p \times K$	Gene weights per program
β	$K \times 1$	Survival coefficients
τ_j	scalar	Gene-specific precision
$q(L_k), q(F_k), q(\beta_k)$	—	EBNM prior (point-normal or point-laplace); hyperparameters estimated via empirical Bayes

The Shared Latent Space

The **same** L drives both:

- Low-rank reconstruction of gene expression ($LF^\top$)
- Cox proportional hazards risk score ($L\beta$)

This forces factors to explain expression variance **and** maximise survival signal simultaneously.

The model has two components sharing the latent loadings L . The first equation is standard matrix factorisation — Y is approximated by L times F -transpose plus gene-specific noise. The second is a Cox proportional hazards model where the same patient loadings L , multiplied by survival coefficients β , give the log hazard ratio. Each column of F defines a gene expression program. Each β_k says whether that program is prognostically relevant.

Dual-Source Objective Function

How the Joint Objective Is Constructed

Genomics Likelihood

$$\mathcal{L}_{\text{genomics}} = \sum_{j=1}^p \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$$

Gaussian reconstruction error — how well $LF^\top$ approximates Y

Survival Likelihood

$$\mathcal{L}_{\text{Cox}} = \sum_{i:\delta_i=1} \left[L_i \beta - \log \sum_{j \in \mathcal{R}_i} \exp(L_j \beta) \right]$$

Cox partial log-likelihood — how well $L\beta$ predicts survival

Combined ELBO

$$\text{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{genomics}}]}_{\text{expression fit}} + \underbrace{E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{D_{\text{KL}}(q \parallel \pi)}_{\text{prior regularisation}}$$

The ELBO **naturally balances** both objectives — no tuning parameter α required. The shared L appears in both terms.

⚠ Gradient Scale Asymmetry

The genomics term sums over p features; the Cox term over n patients. Since $p \gg n$ in genomics data, the genomics gradient dominates — the relative magnitudes are **not normalised** and may be dataset-dependent. Adding a scalar λ to weight $E_q[\mathcal{L}_{\text{Cox}}]$ is an active research direction (see roadmap).

The ELBO is the variational objective we maximise. It decomposes into three intuitive pieces: how well we reconstruct expression, how well we predict survival, and a KL penalty that keeps posteriors close to the prior. The shared L in both likelihood terms is what makes this supervised — the loadings must serve both masters simultaneously.

EBNM & Adaptive Shrinkage

From Penalised Regression to Empirical Bayes

What you already know:

- **Ridge/LASSO** shrink coefficients toward zero with a *fixed* penalty λ
- λ controls how aggressively we regularise
- Chosen by cross-validation — one λ for all coefficients

The empirical Bayes step:

- What if the prior *learned from the data* how much to shrink?
- **EBNM** (Empirical Bayes Normal Means): given noisy observations $x_j = \theta_j + \epsilon_j$, estimate the prior $g(\cdot)$ from the data, then compute posterior $E[\theta_j | x_j, \hat{g}]$
- The shrinkage adapts *per factor* and *per gene*

Two prior families tested:

- **Point-normal**: spike at zero + Gaussian slab; $g(\theta) = \pi_0 \delta_0 + (1 - \pi_0) N(0, \sigma^2)$
- **Point-laplace**: spike at zero + Laplace slab; heavier tails, better preserves large non-zero effects

💡 “The Prior Learns From the Data”

Instead of choosing a penalty strength λ by cross-validation, EBNM estimates the mixing proportion (how sparse?) and the scale (how large are the non-zero effects?) directly from the observed data. Implemented via the **ebnm** R package (Stephens lab, U. Chicago).

i Hyperparameter Specification

- π_0 (sparsity) and σ^2 (slab scale) are estimated by marginal likelihood maximisation at each CAVI step — no user-specified tuning
- τ_j (gene-specific precision): Gamma prior $\tau_j \sim \text{Gamma}(a_0, b_0)$ with weak defaults ($a_0 = b_0 = 10^{-3}$); posterior is closed-form Gamma
- All priors are updated within the ELBO optimisation — the prior adapts to the data at every iteration

Most of you know penalised regression — LASSO, ridge — where lambda controls shrinkage. The key limitation is that lambda is fixed across all coefficients. EBNM generalises this: it estimates the prior from the data using empirical Bayes. The prior has two parts: a spike at zero (for truly zero effects) and a slab (for non-zero effects). The mixing proportion and slab scale are both estimated. Point-normal gives soft thresholding; point-laplace gives heavier tails that preserve larger effects better.

Inference: Factor-Wise CAVI

Coordinate Ascent Variational Inference

Mean-field variational family:

$$q(L, F, \beta, \tau) = \prod_{k=1}^K q(L_{\cdot k}) \prod_{k=1}^K q(F_{\cdot k}) \cdot q(\beta) \cdot q(\tau)$$

Each factor is approximated independently — no posterior correlations across components.

Gauss-Seidel update order (per iteration):

For each factor $k = 1, \dots, K$:

1. **Update** $q(L_k)$ — patient loadings via EBNM
 - Precision: $A_{L,i} = \sum_j \tau_j E[F_{jk}^2] + W_{ii} E[\beta_k^2]$
 - Pseudo-data: $x_i = B_{L,i} / A_{L,i}$, scale: $s_i = 1 / \sqrt{A_{L,i}}$
2. **Update** $q(F_k)$ — gene weights via EBNM (pure genomics)
3. **Update** $q(\beta_k)$ — survival coefficient via EBNM (scalar)
4. Deflate residual R_k before proceeding to $k + 1$

Convergence criteria:

Converged $\iff$ both conditions hold (after > 5 burn-in iterations):

$$\text{mean}_{i,k} |L_{ik}^{(t)} - L_{ik}^{(t-1)}| < 10^{-3} \quad \text{AND} \quad \text{mean}_k |\beta_k^{(t)} - \beta_k^{(t-1)}| < 10^{-3}$$

- Mean over **all** $n \times K$ loading entries (not max) — prevents oscillating orientation boundaries from blocking convergence - Dual criterion: both genomic structure (L) and survival signal (β) must stabilise

Automatic K selection:

- Initialise $K_{\text{max}} = 10$
- Retain factors with either:
 - $\text{PVE} > 1\%$ (non-trivial expression contribution), or
 - $|\hat{\beta}| > 0.05$ (non-negligible survival signal)
- Active factors define K_{eff} ; this is a pragmatic screening rule before future CV-based K selection

CAVI updates one factor at a time in Gauss-Seidel order — this means each update sees the most recent values from all other factors. For each factor k , we update the loadings L , gene weights F , and survival coefficient β , each via an EBNM call. The precision A captures how much information we have about each parameter. The auto-prune starts with 10 factors and lets the data decide how many to keep based on variance explained and survival relevance.

Data Intro — PDAC Cohorts

Independent Validation Across Three Platforms

Dataset	Description	Platform
TCGA_PAAD	The Cancer Genome Atlas — Pancreatic Adenocarcinoma	RNA-seq
CPTAC	Clinical Proteomic Tumor Analysis Consortium	Proteomics
Dijk	Dijk et al. (Dutch PDAC cohort)	RNA-seq
Moffitt_GEO_array	Moffitt Cancer Center (Gene Expression Omnibus)	Microarray
PACA_AU_array	Pancreatic Cancer Australia (ICGC-APGI), array	Microarray
PACA_AU_seq	Pancreatic Cancer Australia (ICGC-APGI), RNA-seq	RNA-seq
Puleo_array	Puleo et al. 2018 (Gastroenterology), multi-centre	Microarray

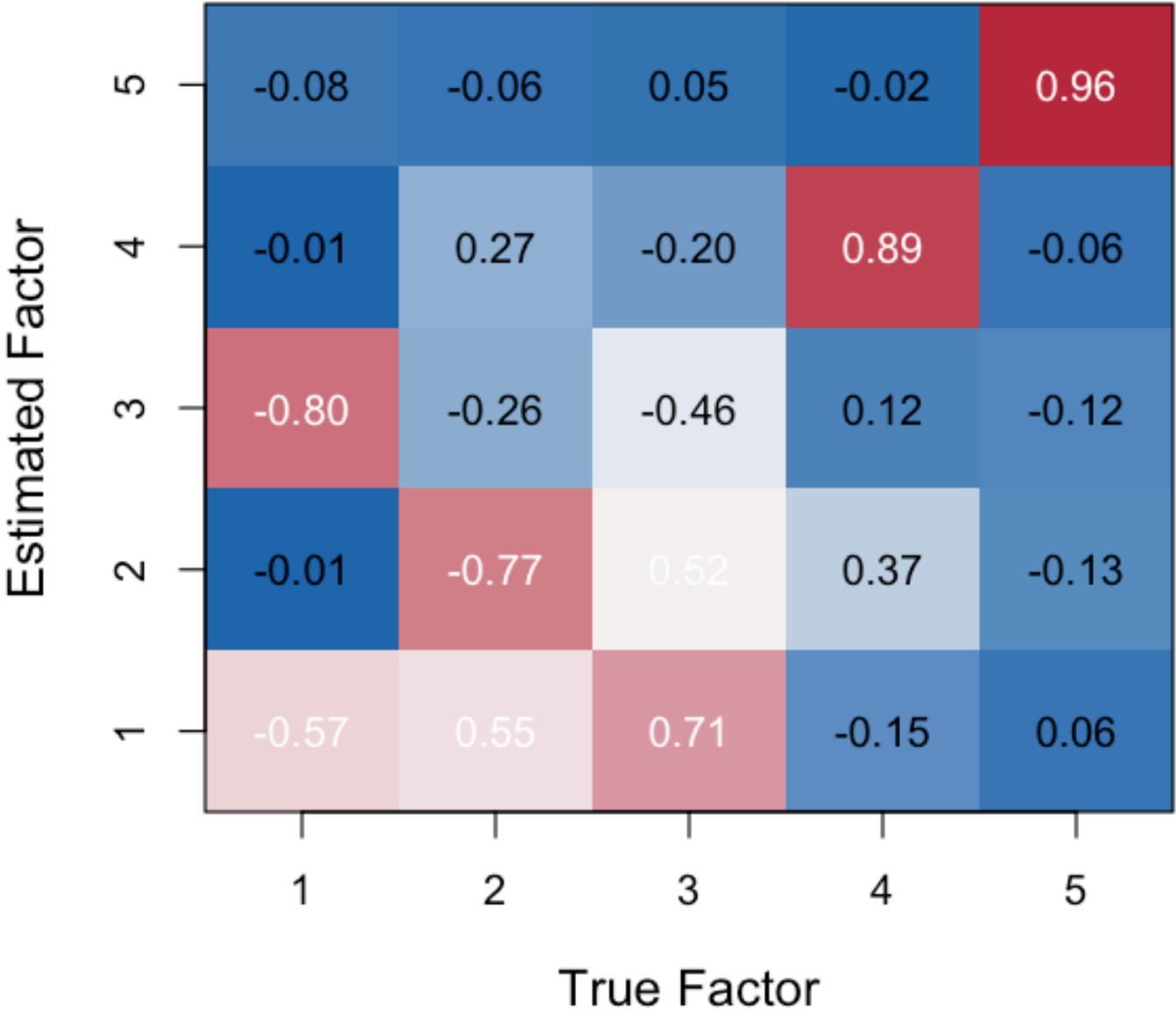
- **RNA-seq** — mRNA transcript counts (TCGA, Dijk, PACA_AU_seq)
- **Microarray** — hybridisation arrays (Moffitt, PACA_AU_array, Puleo)
- **Proteomics** — mass spectrometry protein abundance (CPTAC)
- **Why 7 datasets?** Independent validation across platforms; $n = 52\text{--}288$; top 5,000 most-variable features, column-centred

We applied the model to 7 independent PDAC cohorts spanning three platforms. This is important for showing that the method isn't overfitting to one dataset or one technology. Sample sizes range from 52 (PACA_AU_seq) to 288 (Puleo). The table makes the validation design concrete: we span RNA-seq, microarray, and proteomics, so success here means the method is not tied to one assay. All cohorts use the top 5,000 most-variable features and column centring before initialisation to put the datasets on a comparable footing.

Synthetic Validation

Known Ground Truth Recovery — $n = 200, p = 500, K_{\text{true}} = 5$

Figure 7: |Correlation| Between True and Estimated Loadings



**Figure 3: Estimated Survival Coefficients (95% CI)
(Synthetic (n=250, p=1000, K=5), true beta permutation-aligned)**

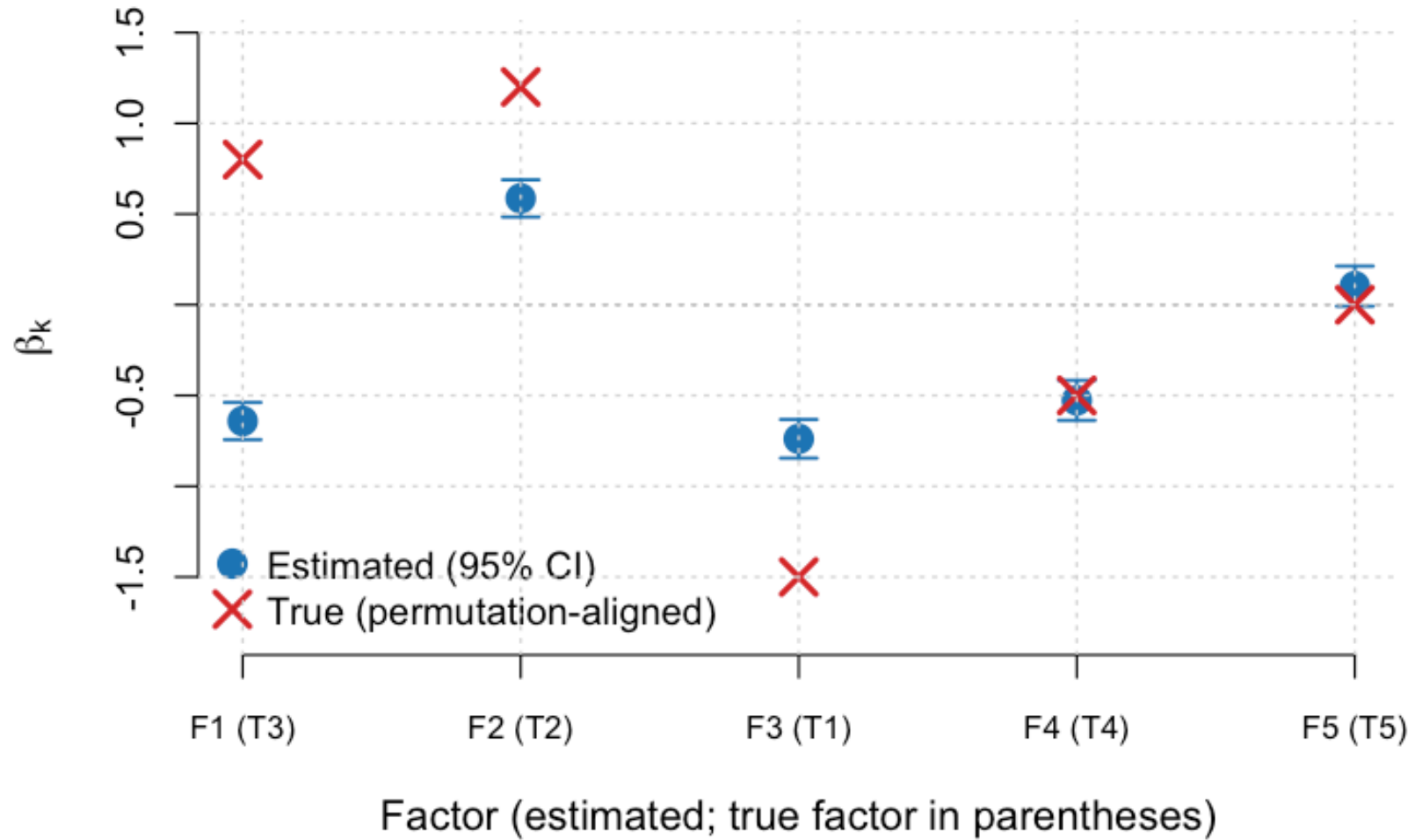


Figure 2: True vs. estimated survival coefficients $\hat{\beta}$

Factor recovery: one-to-one correspondence between true and estimated factors (up to permutation). Factors 1–3 recover cleanly ($r > 0.9$); Factors 4–5 are weaker ($r \approx 0.55$ – 0.67), potentially reflecting genomics gradient dominance compressing smaller survival-relevant programs.

Null factor correctly zeroed: $\hat{\beta}_5 \rightarrow 0$ · **C-index:** Supervised 0.86 vs. PCA 0.83

Proportional Hazards Assumption Verified

Schoenfeld residual test (`cox.zph`): global $p = 0.335$ — no evidence of PH violation. The same test is reported per-dataset in the PDAC results.

Before moving to real data, we validate on synthetic data with known ground truth. The correlation matrix shows clear one-to-one correspondence between true and estimated factors. The beta comparison shows that the model recovers the correct signs and magnitudes, and critically, the null factor (true $\beta=0$) is correctly shrunk to zero. This is the EBNM prior doing its job.

Convergence & Baseline Performance

All Seven Cohorts Converge

Figure 1: Reconstruction RMSE Across CAVI Iterations (Puleo_array)

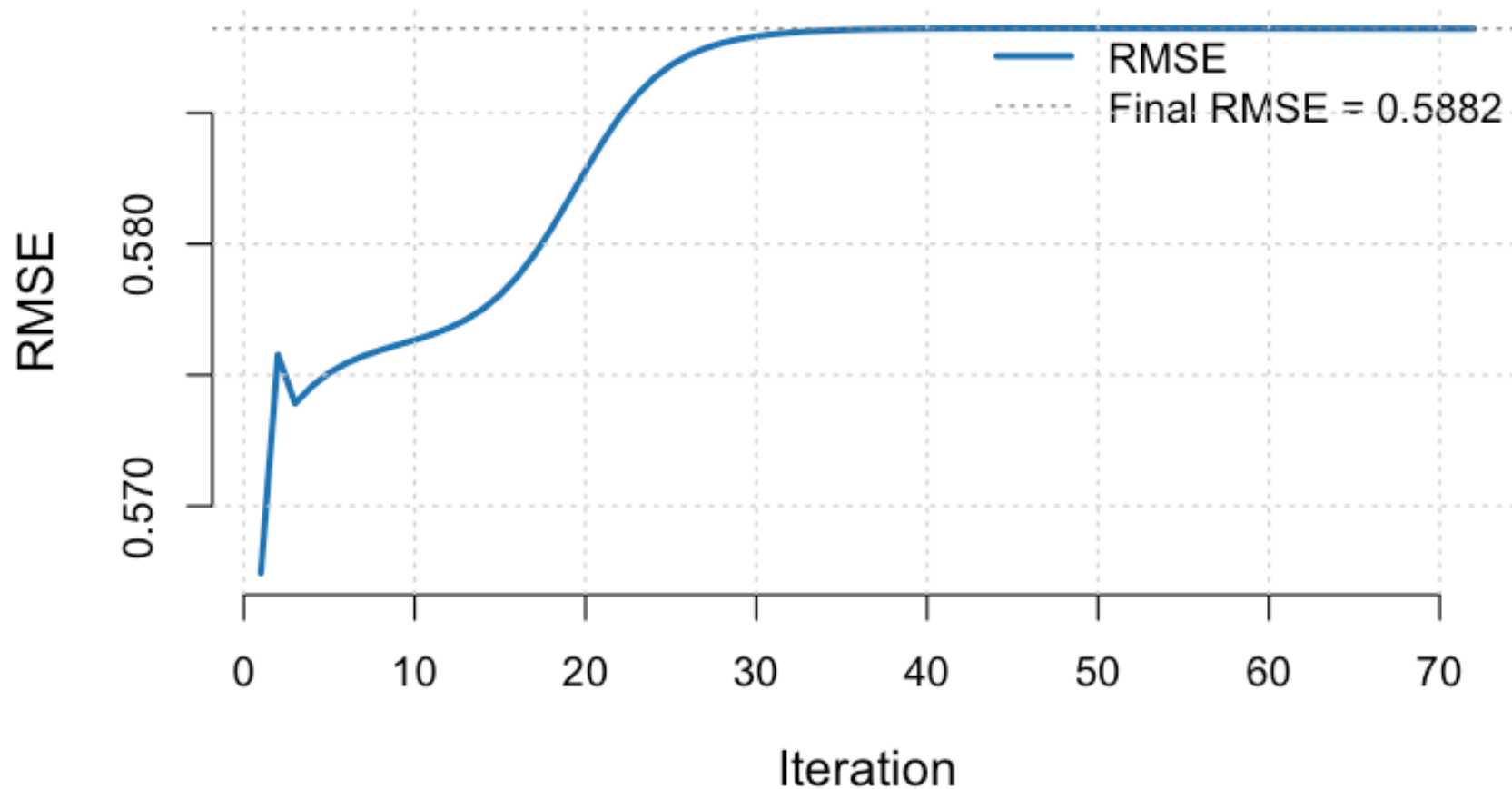


Figure 3: RMSE trace — Puleo array ($n = 288$, converges in 72 iterations)

Figure 1: Reconstruction RMSE Across CAVI Iterations (CPTAC)

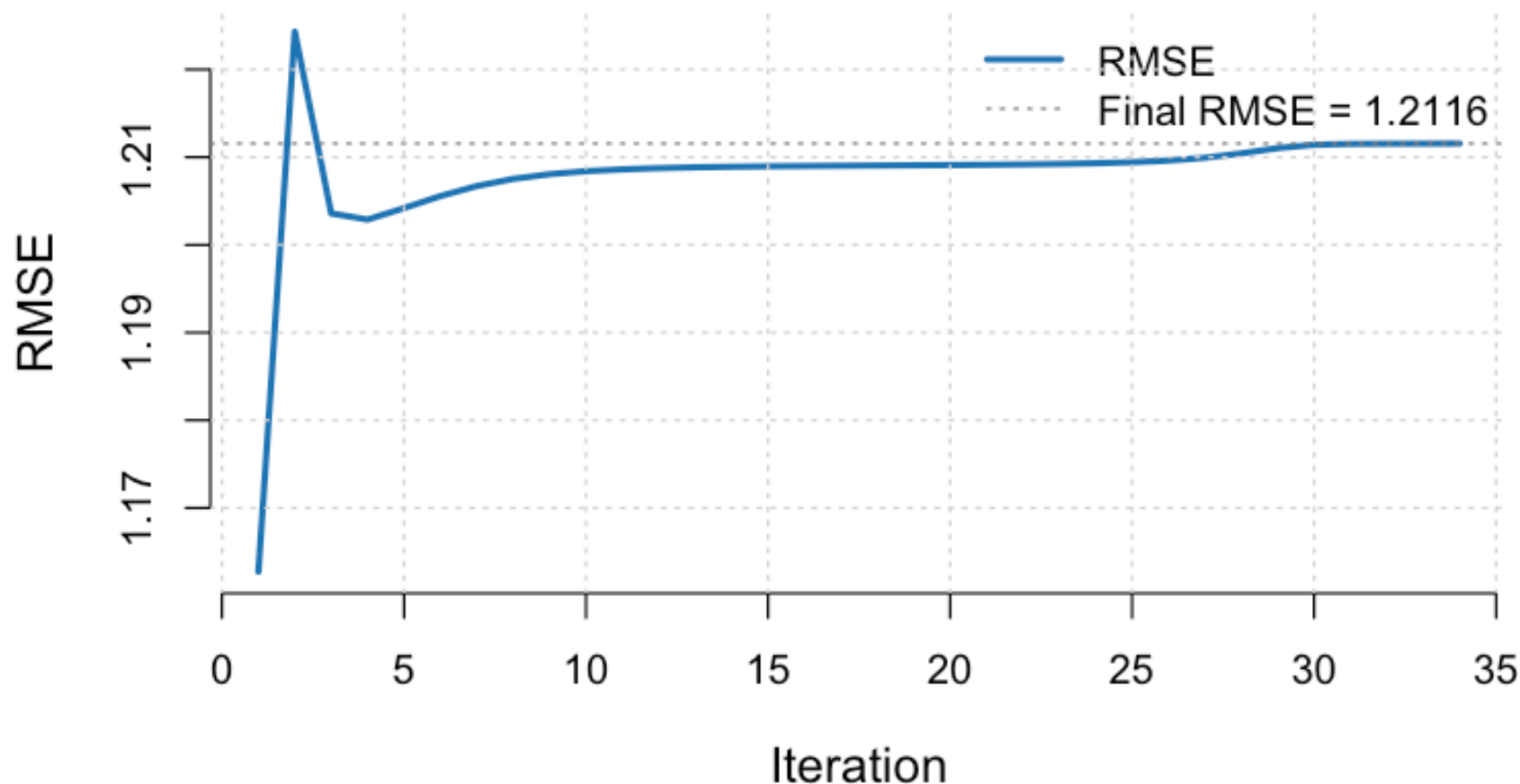


Figure 4: RMSE trace — CPTAC proteomics ($n = 129$, up to 259 iterations in fixed- K run)

- **All 7 cohorts converge** — RMSE stabilises to a plateau in every case
- **Iteration range:** 14 (Moffitt, microarray) to 259 (CPTAC, proteomics)
- Microarray datasets converge fastest; proteomics requires the most iterations (higher platform noise)
- At baseline (point-normal, $K = 5$): supervised C-index $\geq$ PCA in 3/7 cohorts

RMSE converges to a stable plateau in every cohort — the CAVI algorithm is reliably settling. Note that RMSE increases slightly from the PCA initialisation as Bayesian regularisation trades a small amount of reconstruction accuracy for better-regularised, survival-relevant factors; the ELBO (the actual objective) is increasing

Dataset	K_eff	PN K=5	PL K=5	PN K_eff	PL K_eff
CPTAC	8	0.389	0.605	0.635	0.623
Dijk	10	0.381	0.483	0.544*	0.510
Moffitt	10	0.475	0.580*	0.530	0.550
PACA_AU array	8	0.405	0.667	0.738	0.738
PACA_AU seq	9	0.185	0.889	0.741	0.778
Puleo	10	0.396	0.594	0.674	0.649
TCGA	9	0.341	0.707*	0.625*	0.620*

throughout. Puleo is the largest cohort and converges in about 70 iterations. CPTAC proteomics requires up to 259 iterations in the fixed-K configuration, reflecting higher noise in protein abundance measurements. These baseline results use point-normal prior at K=5, which we'll see is actually the worst-case configuration.

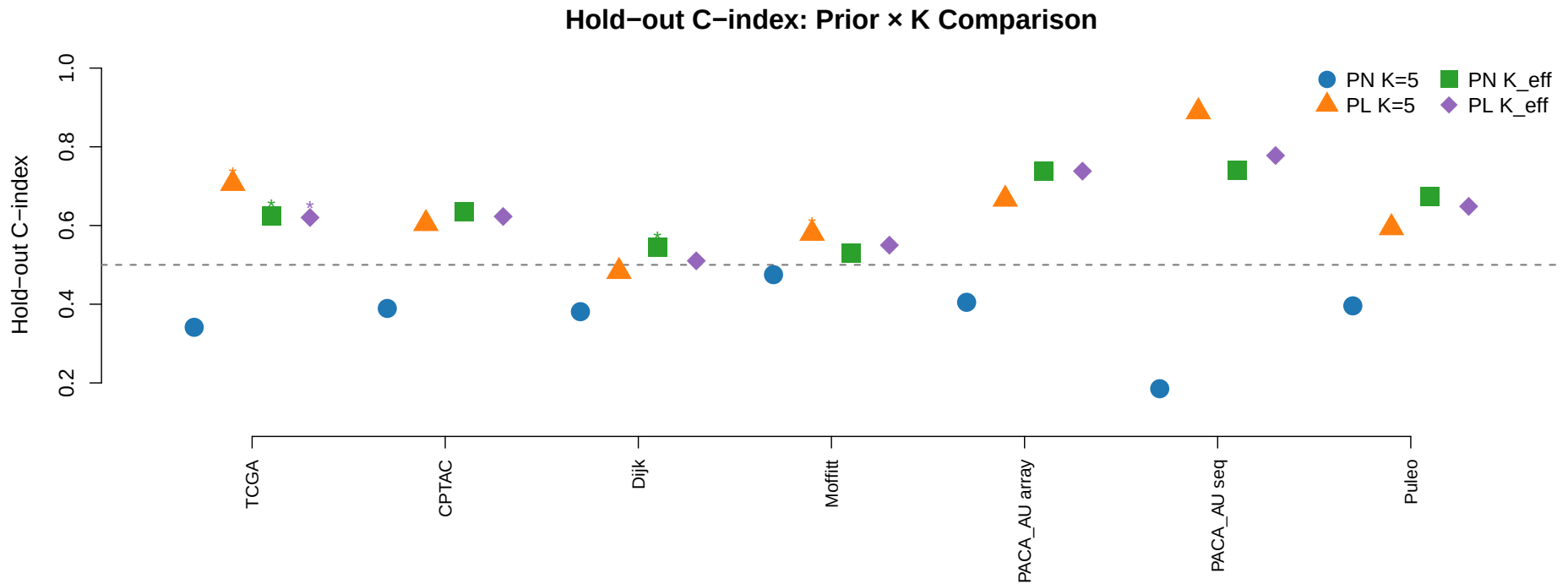
Prior × K Factorial: The Central Result

Hold-Out C-index Across Four Conditions

- = hit 300-iteration limit. PN = point-normal, PL = point-laplace. 80/20 stratified hold-out.
- **Hold-out C-index** — concordance between predicted risk scores and observed survival on 20% held-out patients (random chance = 0.50; higher is better)
- **PN K=5** frequently falls below 0.50 (degenerate solutions); **PL K=5** reliably improves over point-normal
- Moving to **K_eff** raises mean hold-out C-index from 0.507 to 0.640 overall; **PN K_eff** improves on **PN K=5** in all 7 cohorts

This is the central results table. Four conditions per dataset: point-normal or point-laplace prior, at fixed K=5 or data-driven K_eff. Two patterns jump out: point-normal at K=5 often gives C-index below chance (0.19 for PACA_AU_seq, 0.34 for TCGA) — these are degenerate solutions where beta collapses. The cleanest improvement is when we let the model expand beyond K=5: point-normal with K_eff beats its K=5 counterpart in every cohort, and overall average performance is much higher once K is data-adapted.

Prior × K: Scatter Plot



- Each point = one dataset × one condition; dashed line = random chance ($C = 0.50$); four colours = the four prior × K conditions
- **Blue circles (PN K=5):** several datasets fall well below chance — degenerate fits where $\hat{\beta} \rightarrow 0$
- **Moving right (PL → K_eff):** most datasets improve once K is data-adapted, with the biggest rescue coming from PN K=5 to PN K_eff

The scatter plot makes the pattern visual. Blue circles at the bottom are the degenerate point-normal K=5 fits. Orange triangles show the improvement from switching to point-laplace. The biggest pattern, though, is that allowing more factors moves most datasets upward. I would use this slide to emphasise trend and robustness, not to claim that every single condition improves in every cohort.

Key Findings: Prior × K Comparison

Point-laplace preferred at fixed K

- Point-laplace **outperforms** point-normal in **all 7 cohorts** at $K = 5$
- ΔC_{test} from +0.06 (Moffitt) to +0.70 (PACA_AU_seq)
- Point-normal at $K = 5$ degenerates in 4/7 cohorts ($C < 0.4$)
- Heavier tails better capture continuous, non-zero gene weight structure

K selection dominates prior choice

- $K_{\text{eff}} \in \{8, 9, 10\}$ — **always above** $K = 5$; **PN K_{eff}** beats **PN $K=5$** in 7/7 cohorts, while **PL K_{eff}** beats **PL $K=5$** in 4/7
- Mean hold-out C-index increases from **0.507** at fixed $K = 5$ to **0.640** under K-adapted fits
- At K_{eff} : prior advantage is **mixed** (PN wins 4/7)
- Three cohorts saturate at $K_{\text{max}} = 10$

Takeaway: invest in K selection, not prior tuning.

💡 Connecting Back to the Three Design Claims

1. **Data-adaptive priors** → EBNM point-laplace outperforms fixed-penalty (point-normal $K=5$) in all 7 cohorts; $\hat{g}$ is estimated per factor, not shared
2. **Automatic K selection** → K_{eff} (auto-prune via PVE + $|\hat{\beta}|$) beats fixed $K = 5$ in 7/7 cohorts for PN; Factor 9 in Puleo (PVE = 0.91%, $|\hat{\beta}| = 0.36$) would be missed at $K = 5$
3. **Uncertainty quantification** → $\hat{\beta}$ barplots show posterior means with credible intervals; null factor correctly zeroed by EBNM spike

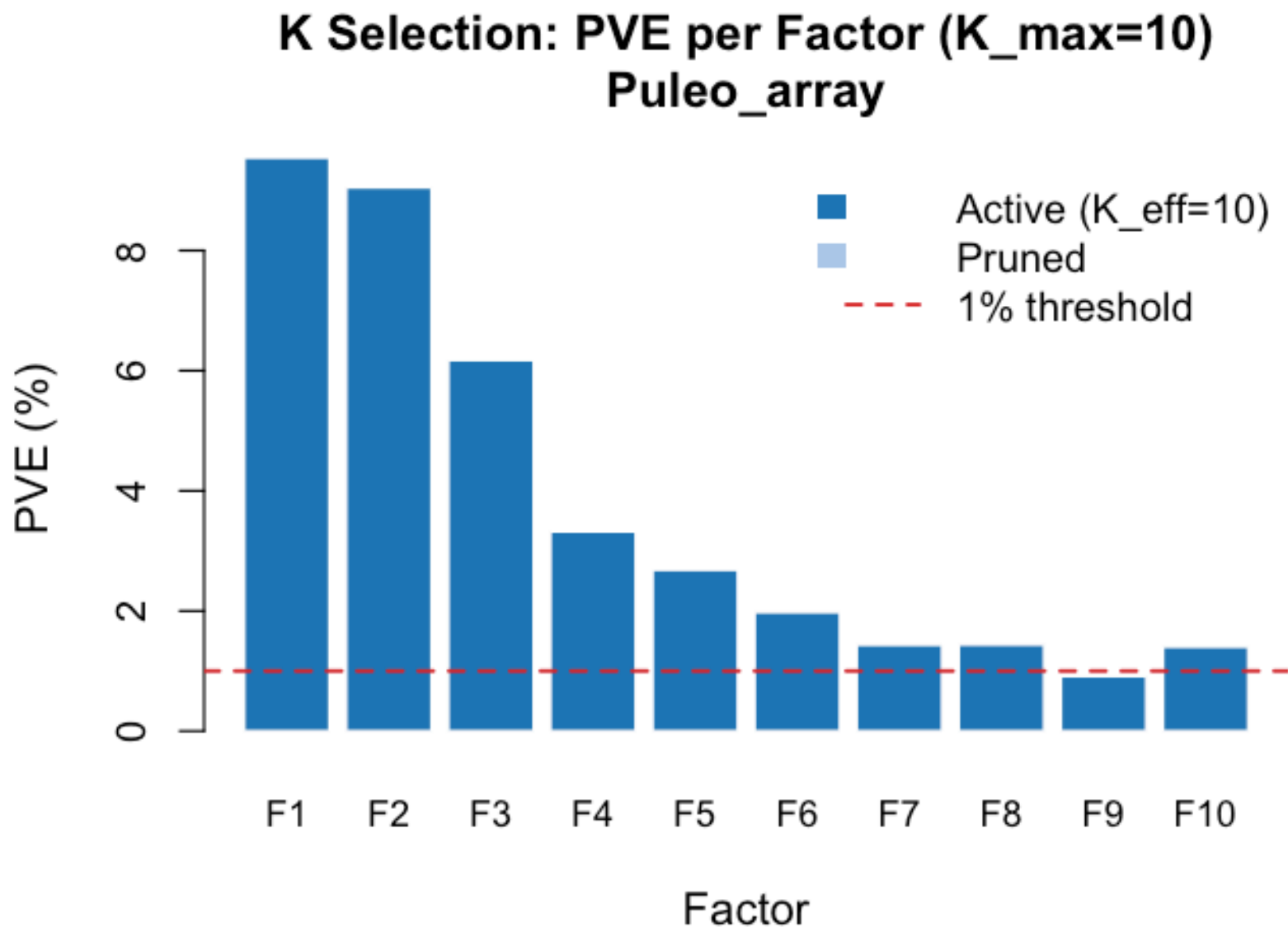


Figure 5: PVE scree — Puleo ($K_{\text{eff}} = 10$)

Two main findings. First, point-laplace is clearly better than point-normal at $K=5$. The Gaussian prior over-shrinks in real data. But the second finding is more important: K selection dominates everything. Once you give the model enough factors, the prior advantage largely disappears. The scree plot shows Puleo needing 10 factors — our initial guess of 5 was way too conservative. Three cohorts actually saturate at our K_{max} of 10, meaning there may be even more structure to discover.

The precise statement I want to make here is that data-adapted K clearly helps overall, not that every single prior-condition pair improves in every cohort.

**Figure 3: Estimated Survival Coefficients (95% CI)
(Puleo_array)**

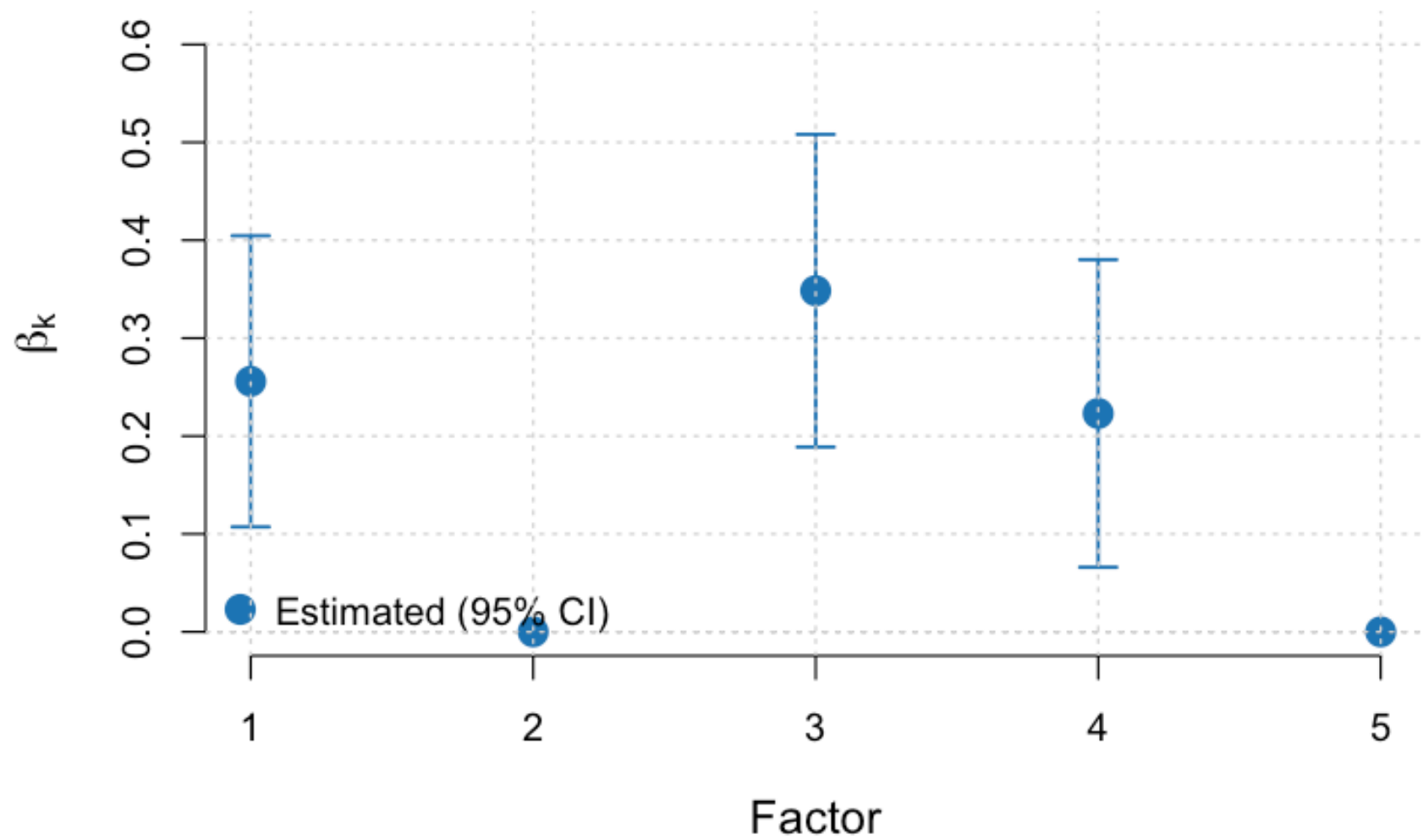


Figure 6: Survival coefficients $\hat{\beta}$ per GEP

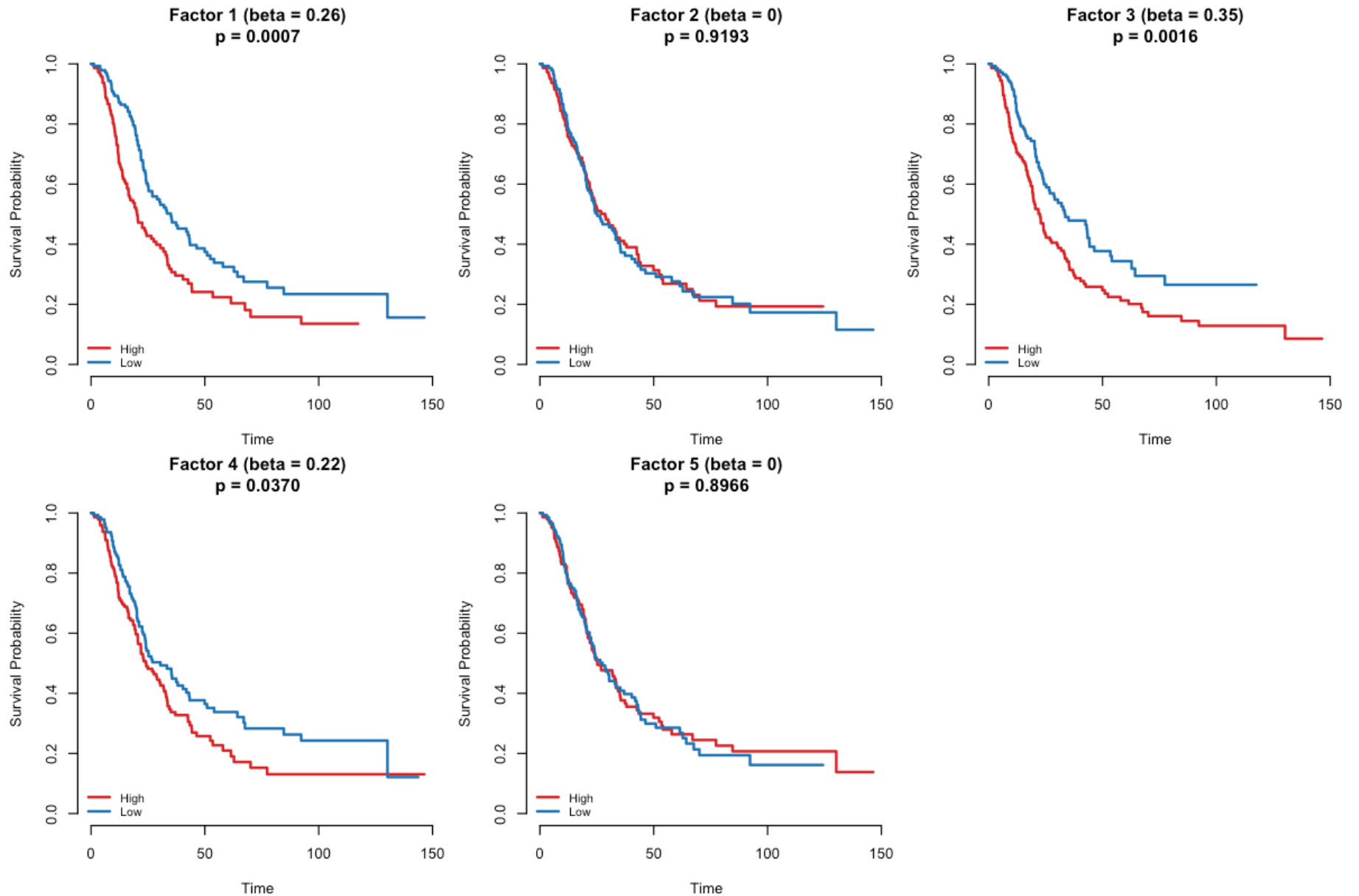


Figure 7: Kaplan-Meier survival curves by GEP activation

- **Factor 2:** highest PVE (9.05%) but $\hat{\beta} \approx 0$ — expression-dominant, survival-neutral (figures show $K = 5$ baseline)
- **Factors 1, 3, 4:** large $|\hat{\beta}|$ — prognostically informative; in the full $K_{\text{eff}} = 10$ model, Factor 9 also emerges ($|\hat{\beta}| = 0.36$, PVE = 0.91%)
- The model **separates prognostic from non-prognostic** structure — PCA would test all factors indiscriminately
- $K_{\text{eff}} = 10$ via auto-prune (4 more factors than fixed $K = 5$)

Puleo is our best exemplar — largest cohort, clearest structure. The beta barplot shows factor 2 is shrunk to zero despite being the single largest expression axis. A post-hoc approach would have tested it alongside the genuinely prognostic factors, diluting statistical power. The KM curves show clear patient stratification by GEP activation. This is what supervised factorisation buys you.

Figure 3: Estimated Survival Coefficients (95% CI)
(TCGA_PAAD)

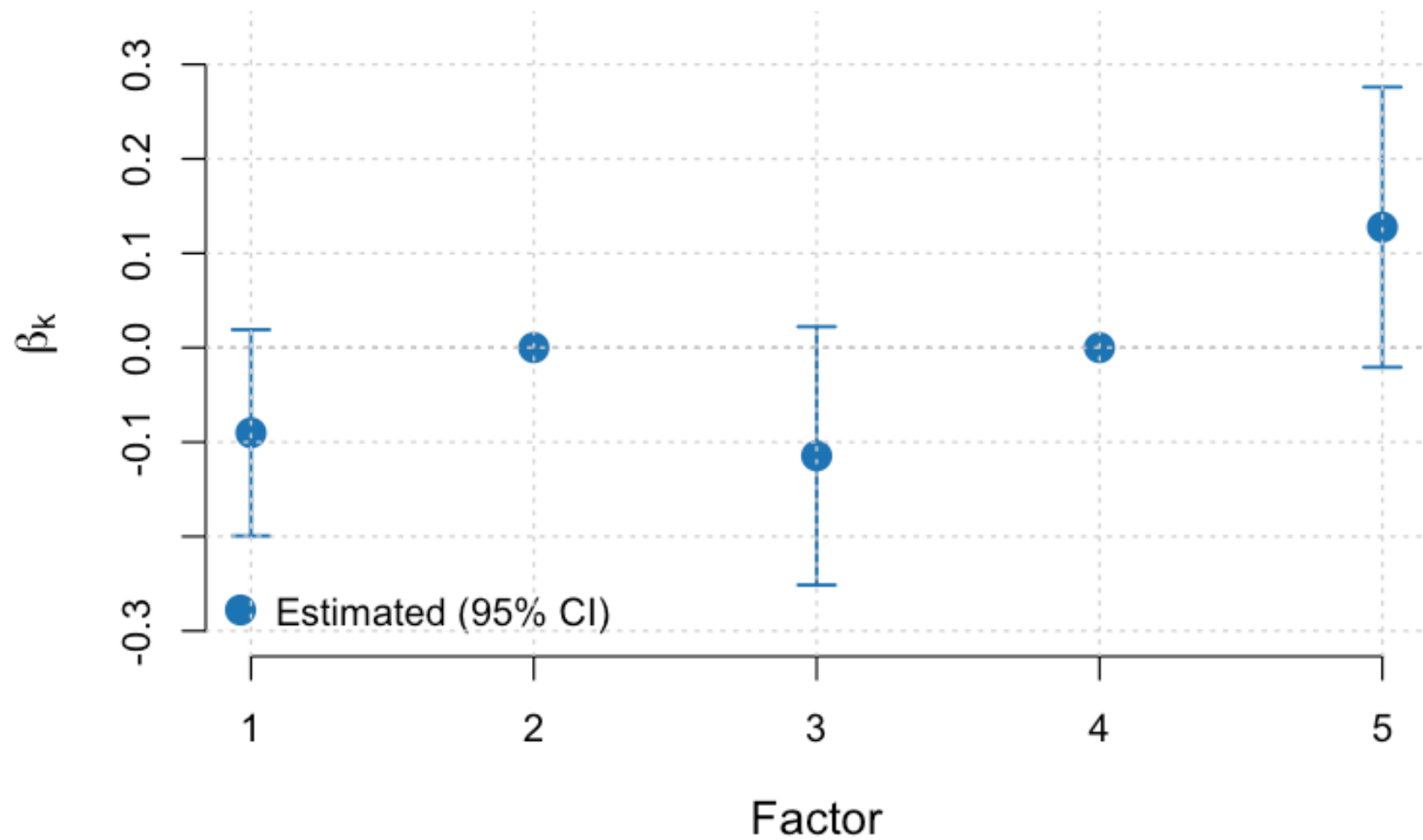


Figure 8: Survival coefficients $\hat{\beta}$ per GEP

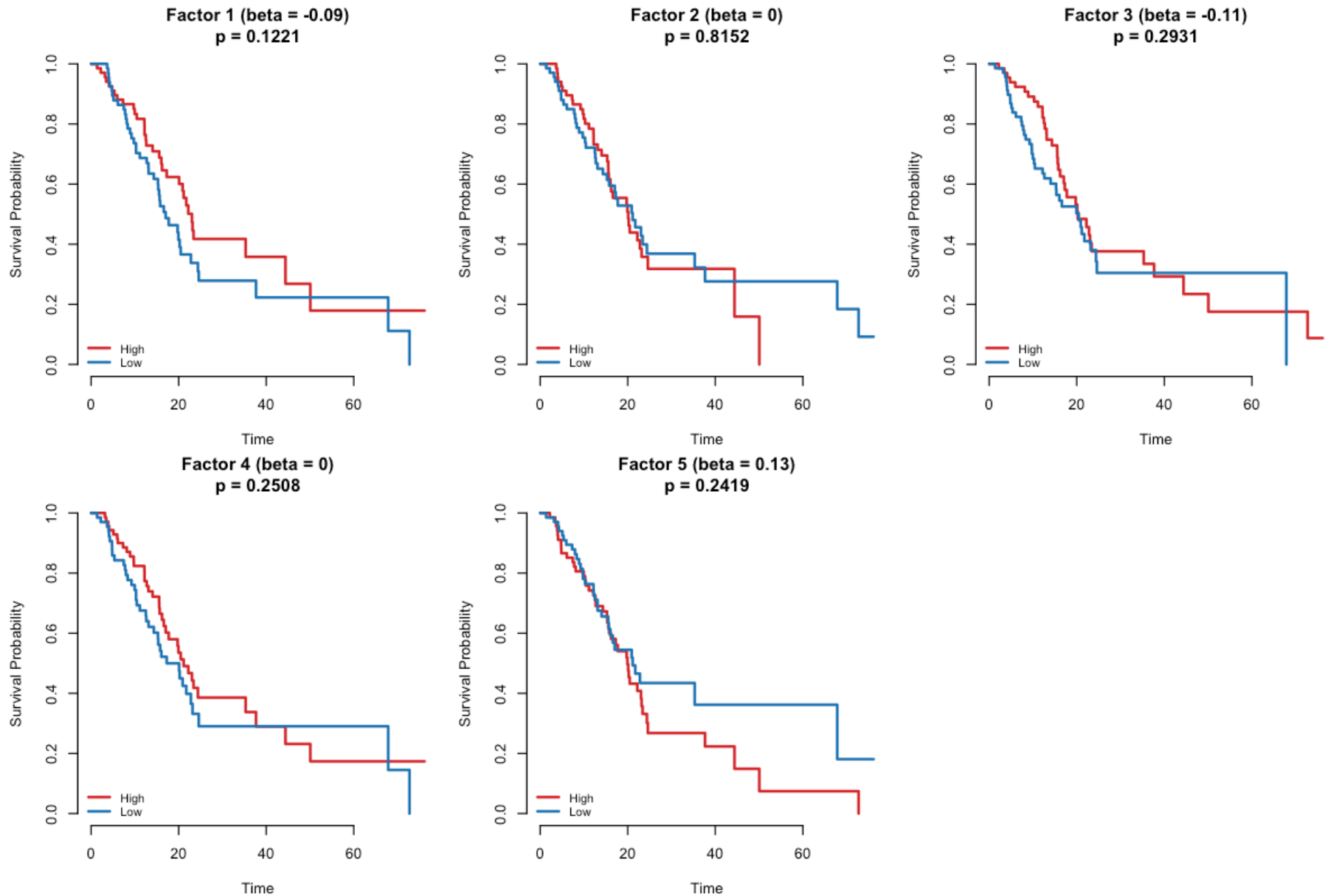


Figure 9: Kaplan-Meier survival curves by GEP activation

- **TCGA_PAAD** ($n = 144$, RNA-seq) — the benchmark PDAC dataset
- $K_{\text{eff}} = 9$; largest K_{eff} improvement: $\Delta C_{\text{test}} = +0.28$ over $K = 5$ baseline
- Consistent with Puleo: model identifies both prognostic and expression-only GEPs
- Patient stratification trends visible in Kaplan-Meier curves; $n = 144$ limits per-factor statistical power

Dataset	C (PCA)	Best Condition	K	C (SBMF)	ΔC
CPTAC	0.703	PN K_eff	8	0.635	-0.068
Dijk	0.610	PN K_eff	10	0.544	-0.066
Moffitt	0.571	PL K=5	5	0.580	0.009
PACA_AU array	0.653	PN K_eff	8	0.738	0.085
PACA_AU seq	0.776	PL K=5	5	0.889	0.113
Puleo	0.664	PN K_eff	10	0.674	0.010
TCGA	0.640	PL K=5	5	0.707	0.067

TCGA is the dataset most people in the room will have heard of. It shows the same pattern as Puleo: some factors are prognostic, others are expression-only. The K_eff improvement here is dramatic — going from K=5 to K=9 improves hold-out C-index by 0.28. This cohort is also where we see the most dramatic failure of point-normal at K=5 (C-index of 0.34 — below random).

Hold-Out Prediction

Generalisation to Unseen Patients

- SBMF beats PCA baseline on held-out patients in 5/7 cohorts
- Improvements most reliable when test set is large enough ($n_{\text{test}} \geq 20$)
- **Caveat:** small test sets ($n \leq 15$) produce noisy C-index estimates

On held-out data, SBMF generalises better than PCA in 5 of 7 cohorts. The improvements are most reliable in larger test sets. PACA_AU_seq has only about 10 test patients — that's too few for a reliable C-index estimate. This table is there to answer the practical question: does the method help on new patients, not just during training? The answer is usually yes, but I should also be upfront that the smallest cohorts produce noisy estimates.

Future Directions & Summary

Next Steps

1. **Gene set enrichment** — fgsea / Enrichr on top-loaded genes per GEP (MSigDB, KEGG, Reactome)
2. **CV-based K selection** — cross-validated C-index on Longleaf HPC ($K_{\text{max}} \geq 15$)
3. **Multi-dataset shared factors** — identify GEPs replicating across cohorts
4. **Full ELBO as model selection** — compare models by variational lower bound
5. **Manuscript preparation**

Summary

1. **SBMF** jointly learns gene expression programs and survival associations via Bayesian inference
2. **K selection is the dominant lever** — $K_{\text{eff}} \in \{8, 9, 10\}$ vs. fixed $K = 5$
3. **Point-laplace preferred** when K is fixed at 5
4. **Generalises** to held-out patients across 3 platforms (5/7 cohorts)
5. **Codebase:** 139/139 tests passing, modular architecture

The most pressing next step is gene set enrichment to give biological names to the statistical factors. Second, proper cross-validated K selection on our HPC cluster with K_max of 15 or higher. Third, finding which GEPs replicate across independent cohorts would be the strongest biological validation. To summarise: the method works, converges reliably across platforms, and the main insight is that getting K right matters more than prior selection.

Discussion

Questions, suggestions, and feedback welcome!

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Open the floor for discussion. Happy to go deeper on any part of the methods, results, or future plans. I have backup slides with additional per-cohort results, ELBO traces, and technical details on EBNM.

Pooled RNA-seq Analysis

Cross-Cohort Validation: TCGA + Dijk + PACA_AU_seq

**Figure 1: Reconstruction RMSE Across CAVI Iterations
(Pooled RNA-seq: TCGA_PAAD+Dijk+PACA_AU_seq)**

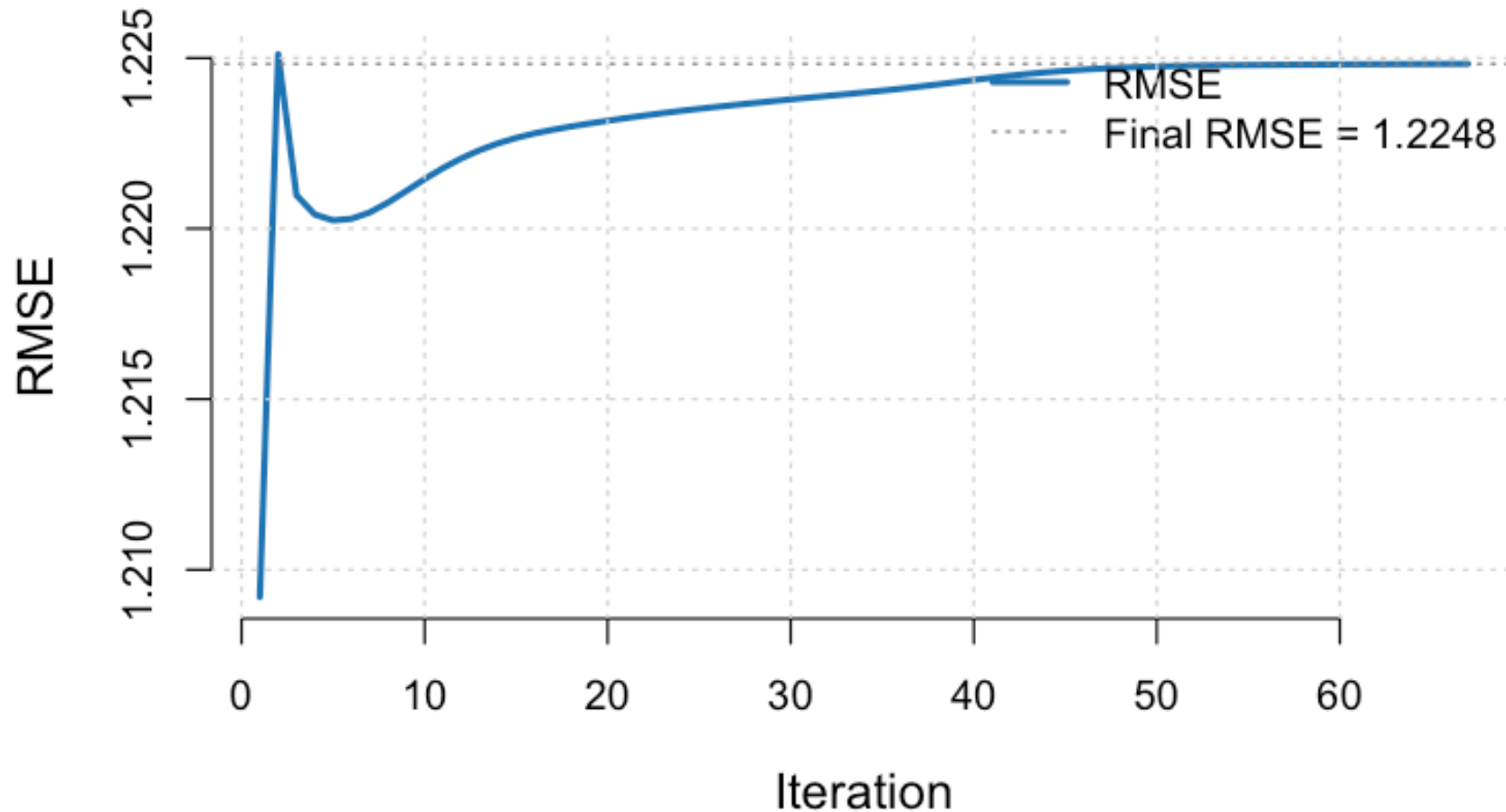


Figure 10: RMSE convergence — pooled RNA-seq ($n = 286$)

**Figure 3: Estimated Survival Coefficients (95% CI)
(Pooled RNA-seq: TCGA_PAAD+Dijk+PACA_AU_seq)**

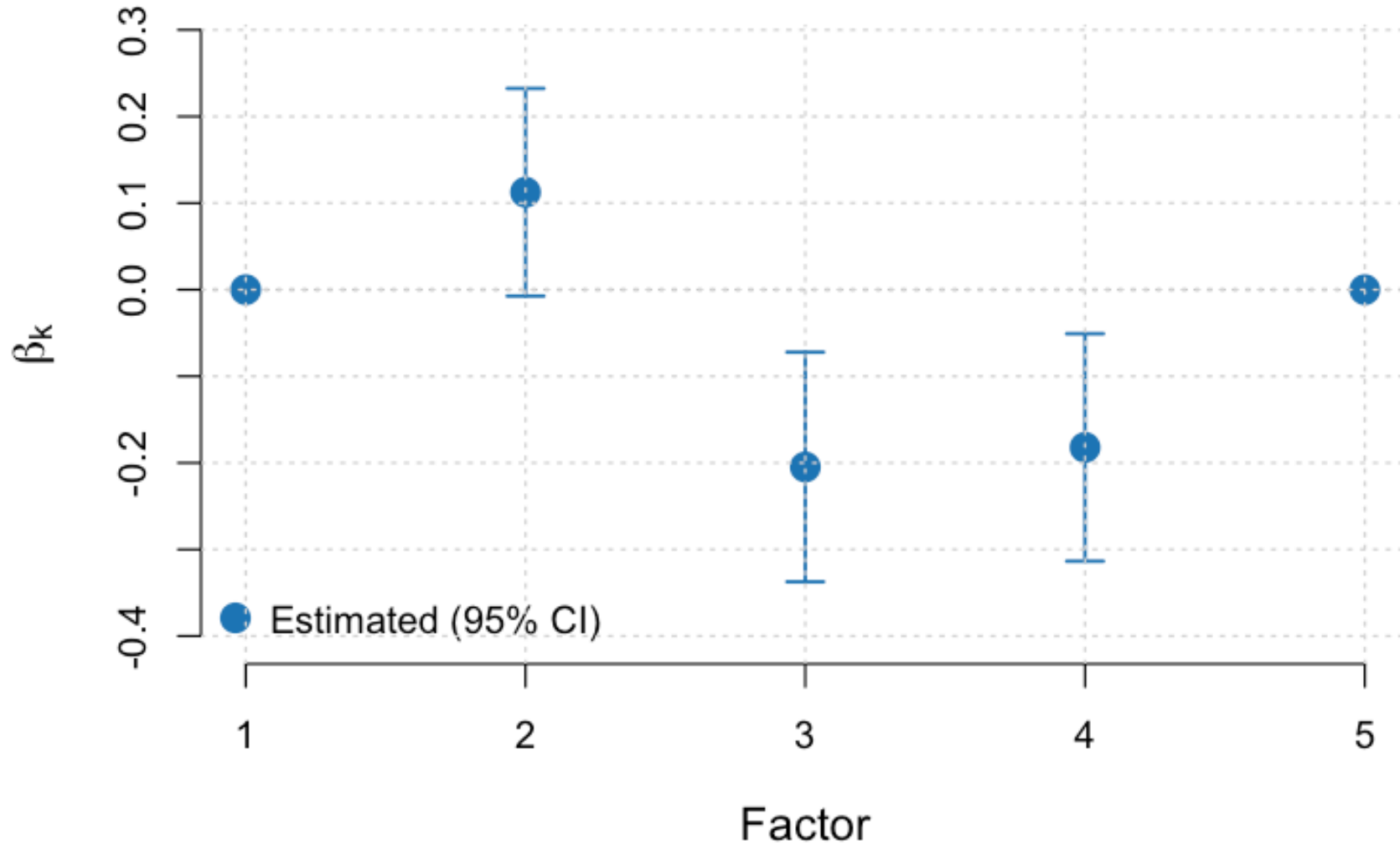


Figure 11: Survival coefficients $\hat{\beta}$ — pooled

- **Pooled:** $n = 286$ patients, 1,573 common genes (intersection across 3 RNA-seq cohorts)
- Converges in 67 iterations
- Tests whether gene programs replicate when patients from different studies are combined

Use this backup slide if someone asks whether the model can find shared signal across studies instead of only within one cohort. The key message is that a pooled RNA-seq fit is feasible, converges cleanly, and gives a first pass at cross-cohort replication once we restrict to genes shared by all three studies.

EBNM Deep Dive

Empirical Bayes Normal Means — Full Detail

Each CAVI update reduces to an EBNM sub-problem:

Normal-means form

- Observed pseudo-data = signal + noise
- Pseudo-data term: $\mathbf{x}_j = \mathbf{B}_j / \mathbf{A}_j$
- Scale term: $\mathbf{s}_j = \mathbf{A}_j^{-1/2}$
- Noise model: $\epsilon_{j,k} \sim \text{Normal}(0, \mathbf{s}_j^2)$

What EBNM estimates

- Learns the prior $\mathbf{g}$ by maximizing the marginal likelihood over candidate priors
- Returns a posterior mean for each coordinate
- Returns a posterior variance for each coordinate
- Critical second moment = variance + mean squared

This prevents overfitting to uncertain parameter estimates and gives the error-in-variables correction used in the SBMF updates.

This is the technical backup for anyone who wants to know what the EBNM calls are actually doing. I would focus on the pseudo-data view, then end on the second-moment correction because that is the main reason the Bayesian update is safer than plugging in posterior means alone.

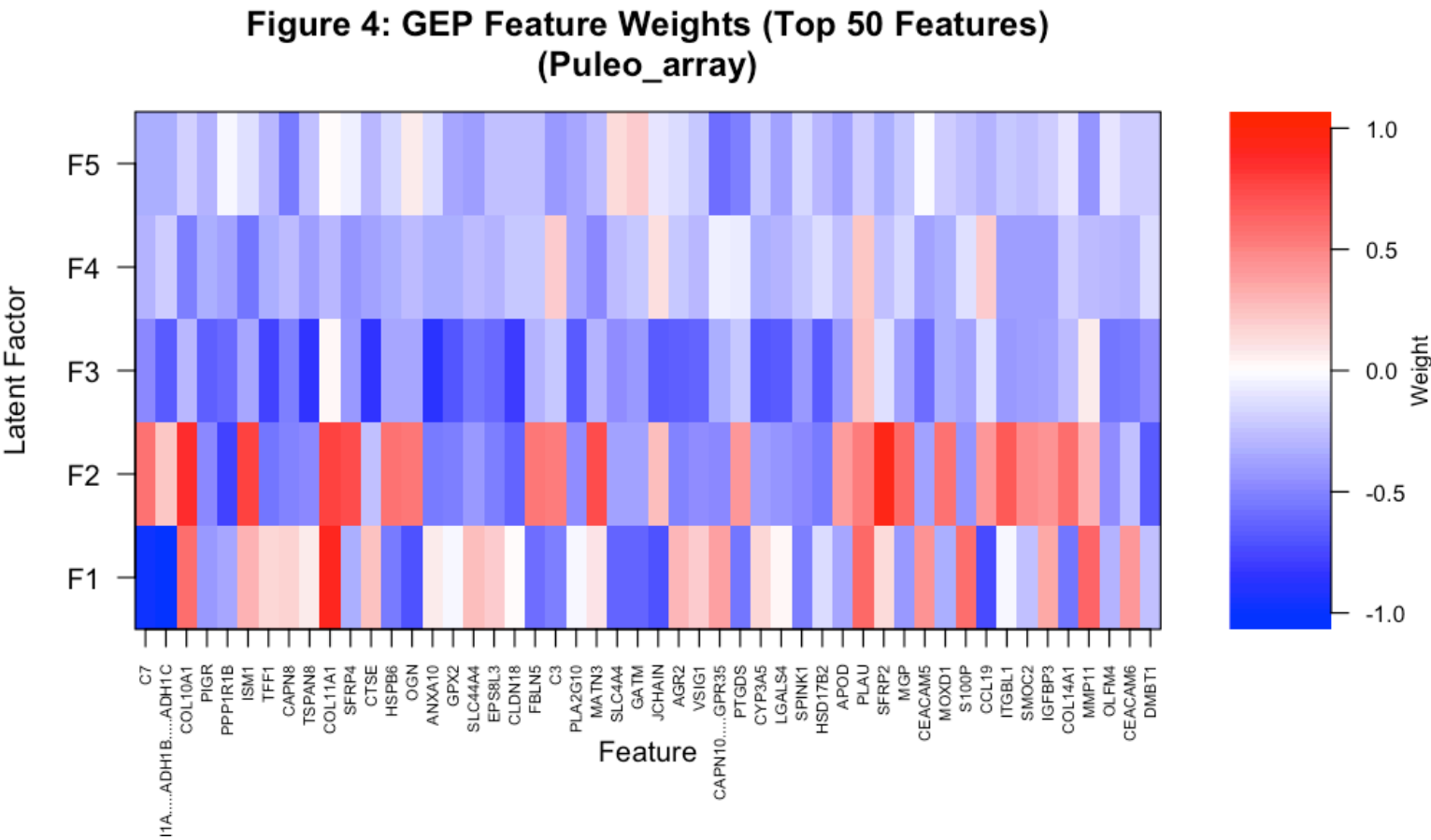


Figure 12: Gene expression program heatmap — patient loadings (rows) × GEPs (columns)

Each column is a gene expression program. Distinct patient groups emerge from the loading patterns.

Use this slide if someone wants a visual sense of patient heterogeneity rather than a single performance number. The point is that the latent programs are not abstract artifacts: they produce structured loading patterns that group patients in biologically interpretable ways.

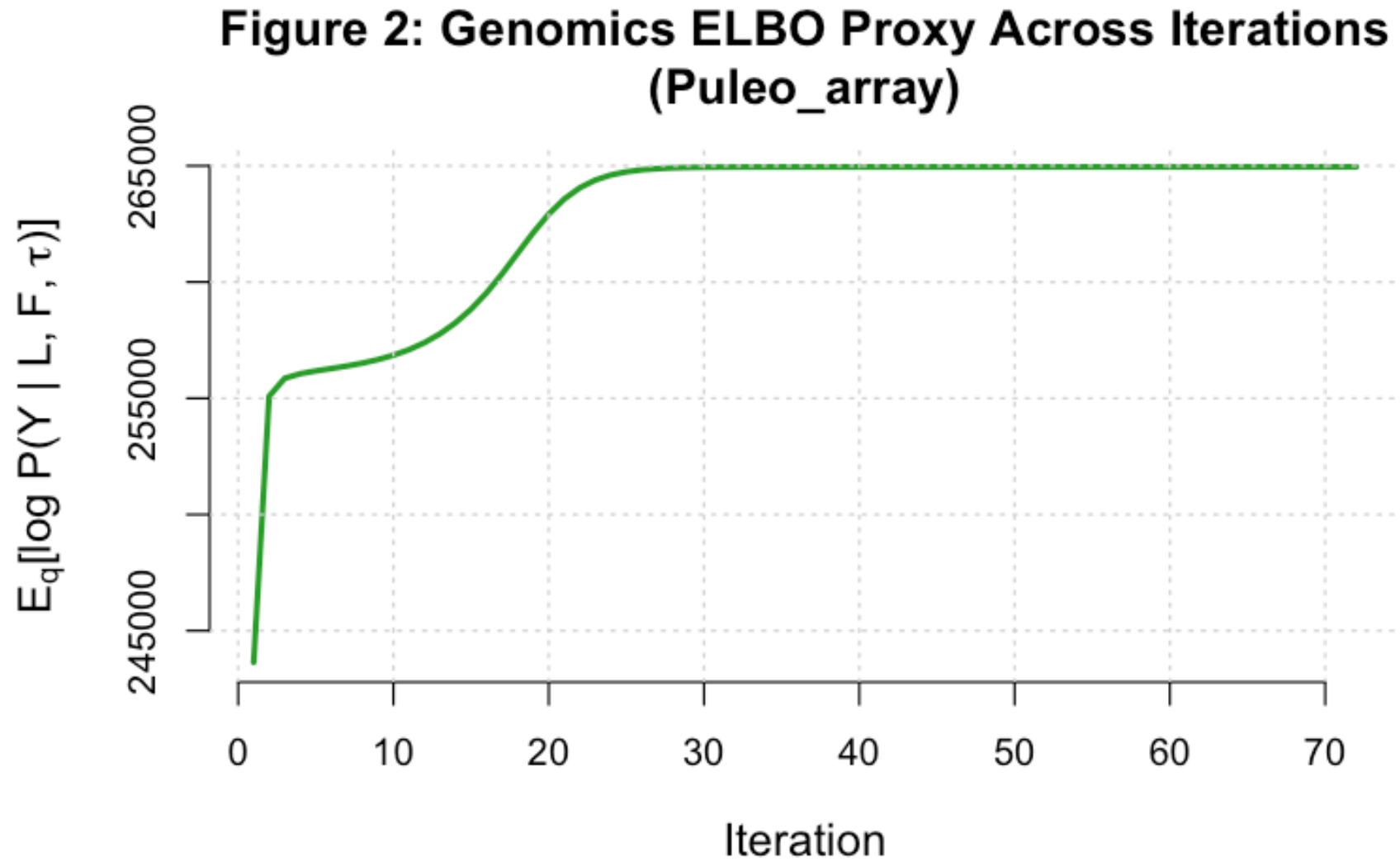


Figure 13: ELBO trace — Puleo array

**Figure 2: Genomics ELBO Proxy Across Iterations
(TCGA_PAAD)**

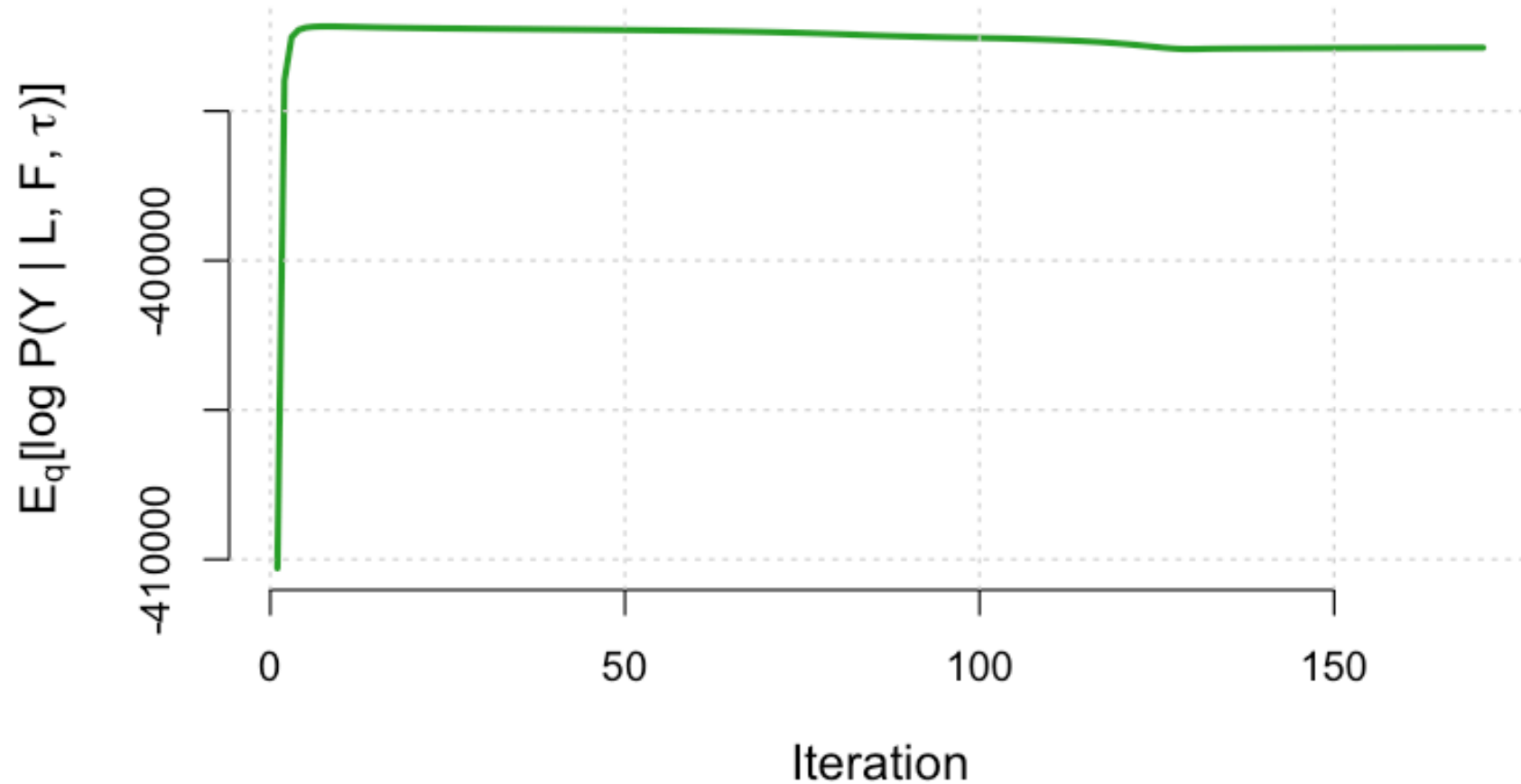


Figure 14: ELBO trace — TCGA_PAAD

ELBO (variational lower bound) increases monotonically, confirming valid CAVI convergence.

This backup addresses the natural question of whether the optimiser is behaving properly. If asked, I would say the traces are a sanity check that the variational objective is moving in the right direction, complementing the RMSE plots in the main talk.

Figure 8: Estimated Feature-Specific Noise Precision (Puleo_array)

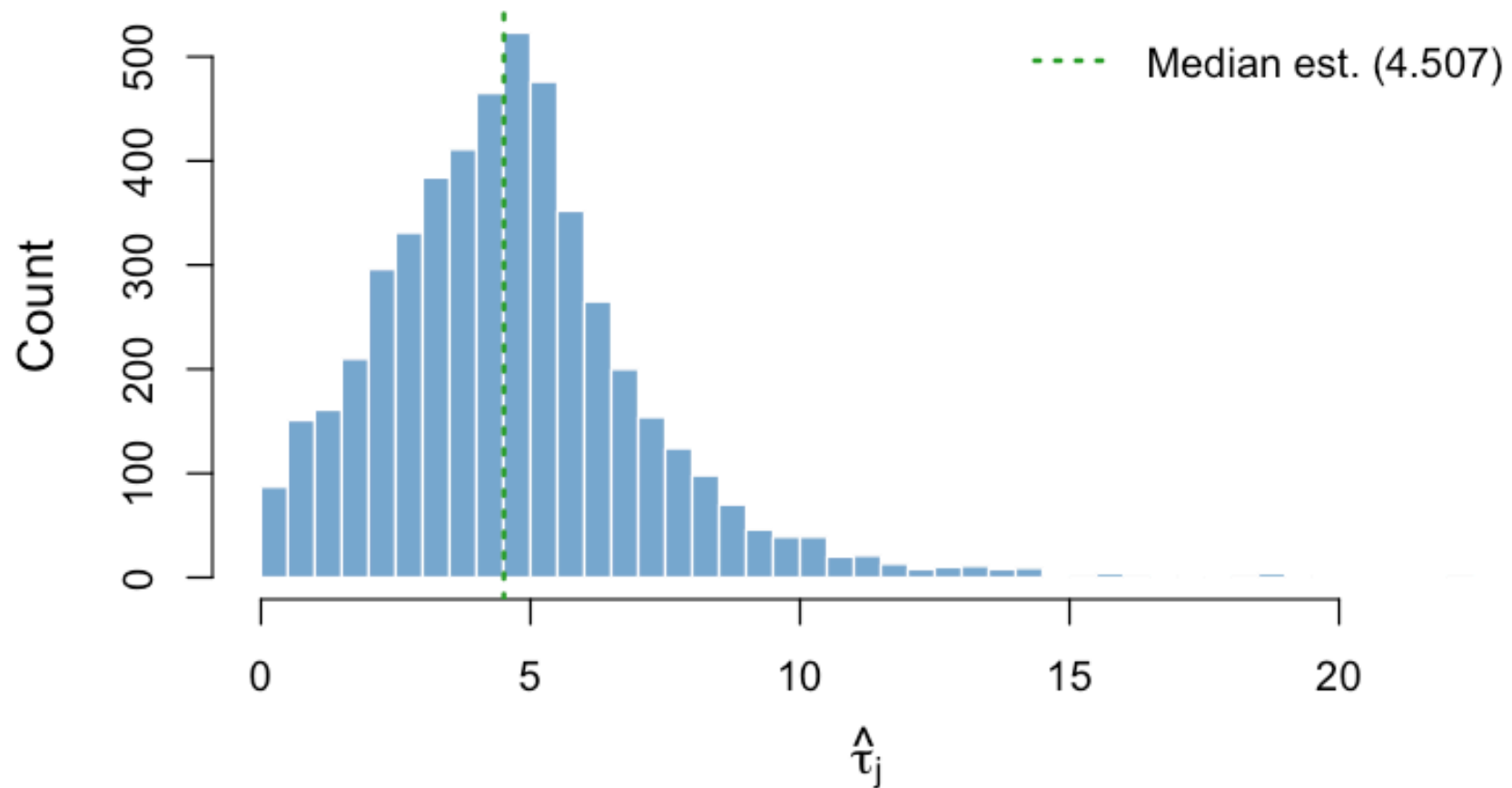


Figure 15: Gene-specific precision $\hat{\tau}_j$ — Puleo array

Figure 8: Estimated Feature-Specific Noise Precision (CPTAC)

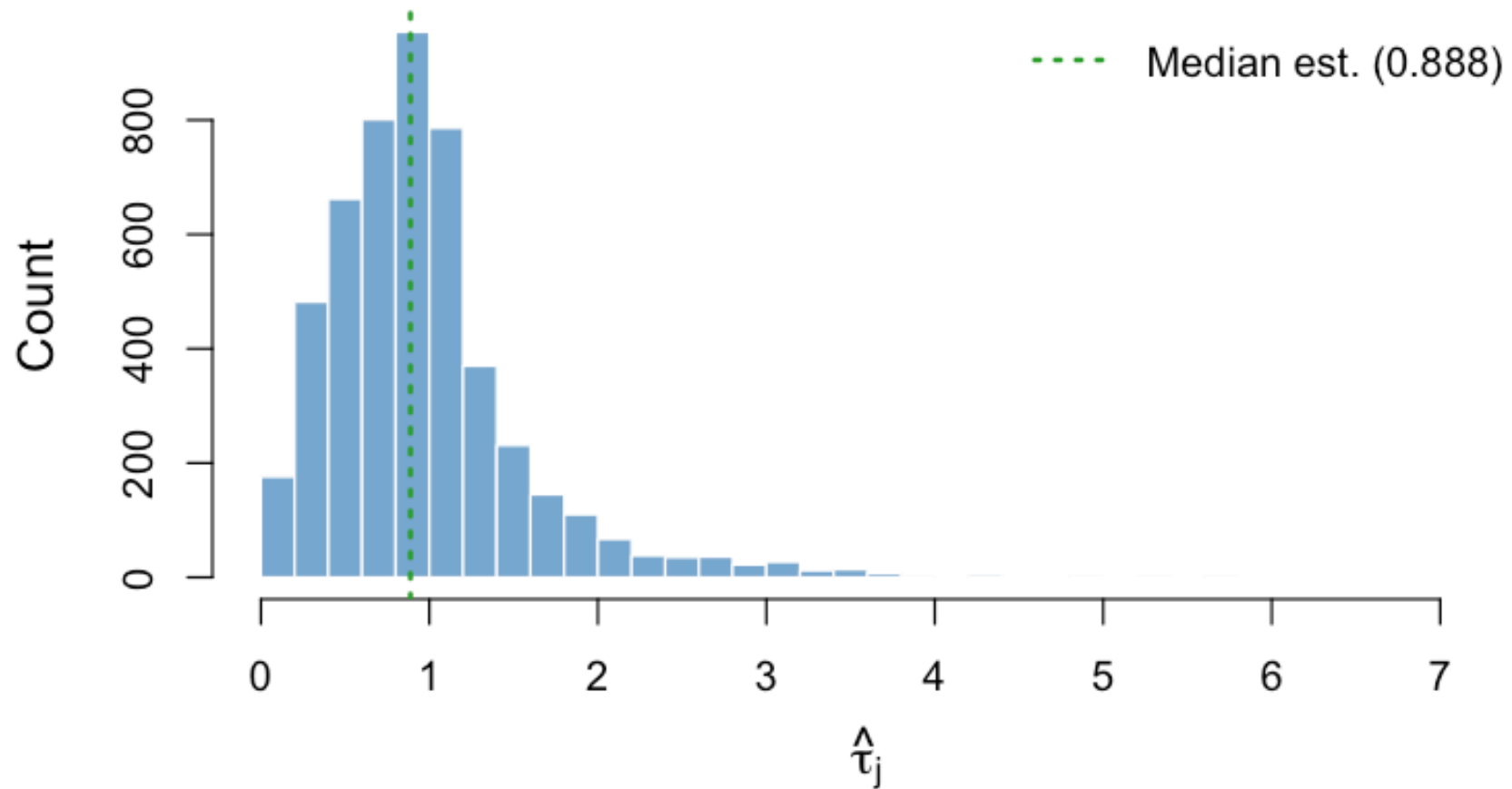


Figure 16: Gene-specific precision $\hat{\tau}_j$ — CPTAC

Heteroscedastic noise model: each gene has its own precision estimate. Some genes are precisely measured (high τ), others are noisy (low τ).

Use this slide if someone asks why the model includes gene-specific precision. The takeaway is that heteroscedasticity is visible in real data, especially across platforms, and estimating τ_j lets the model down-weight noisier features instead of treating every gene as equally reliable.

Figure 3: Estimated Survival Coefficients (95% CI)
(Dijk)

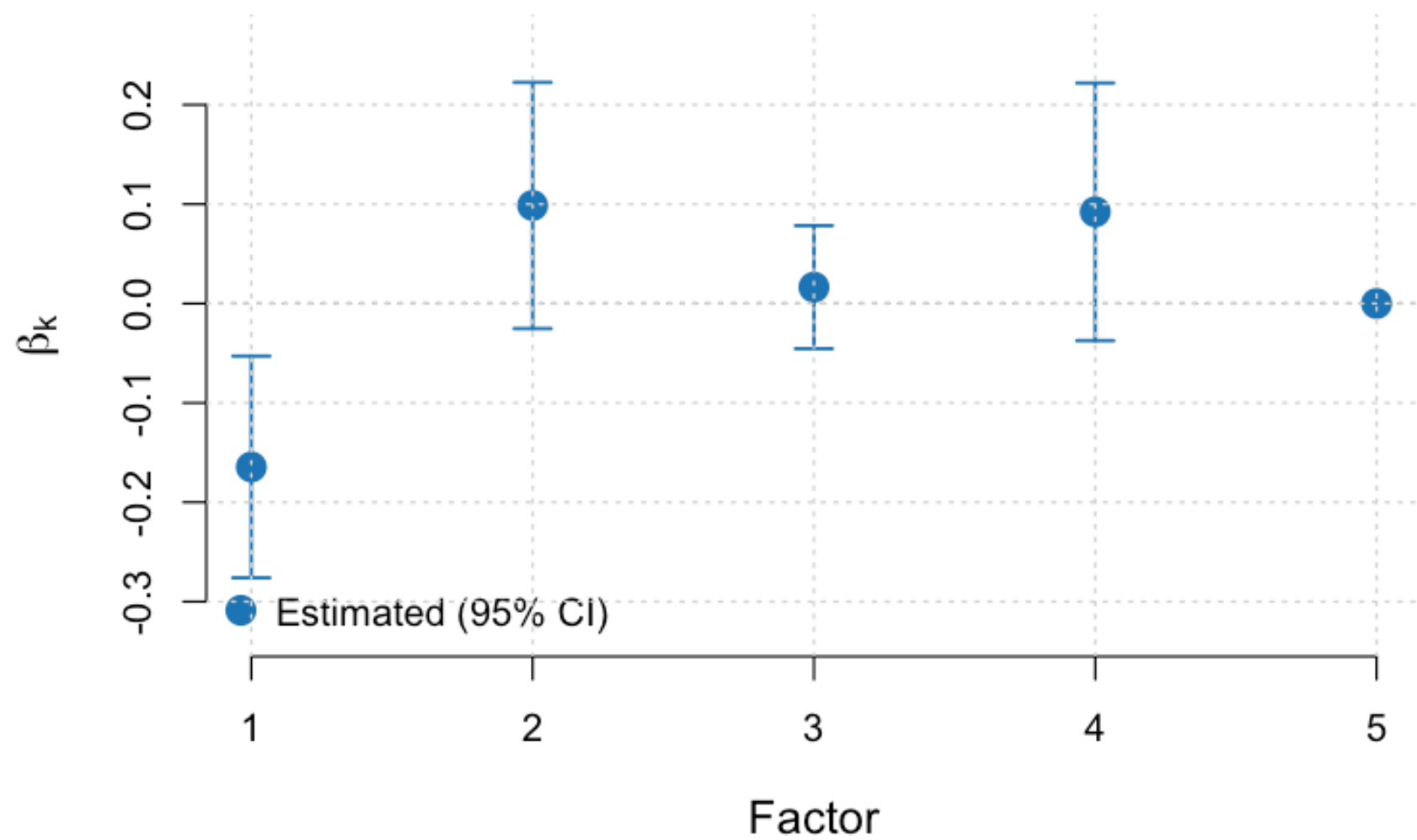


Figure 17: Dijk

**Figure 3: Estimated Survival Coefficients (95% CI)
(PACA_AU_array)**

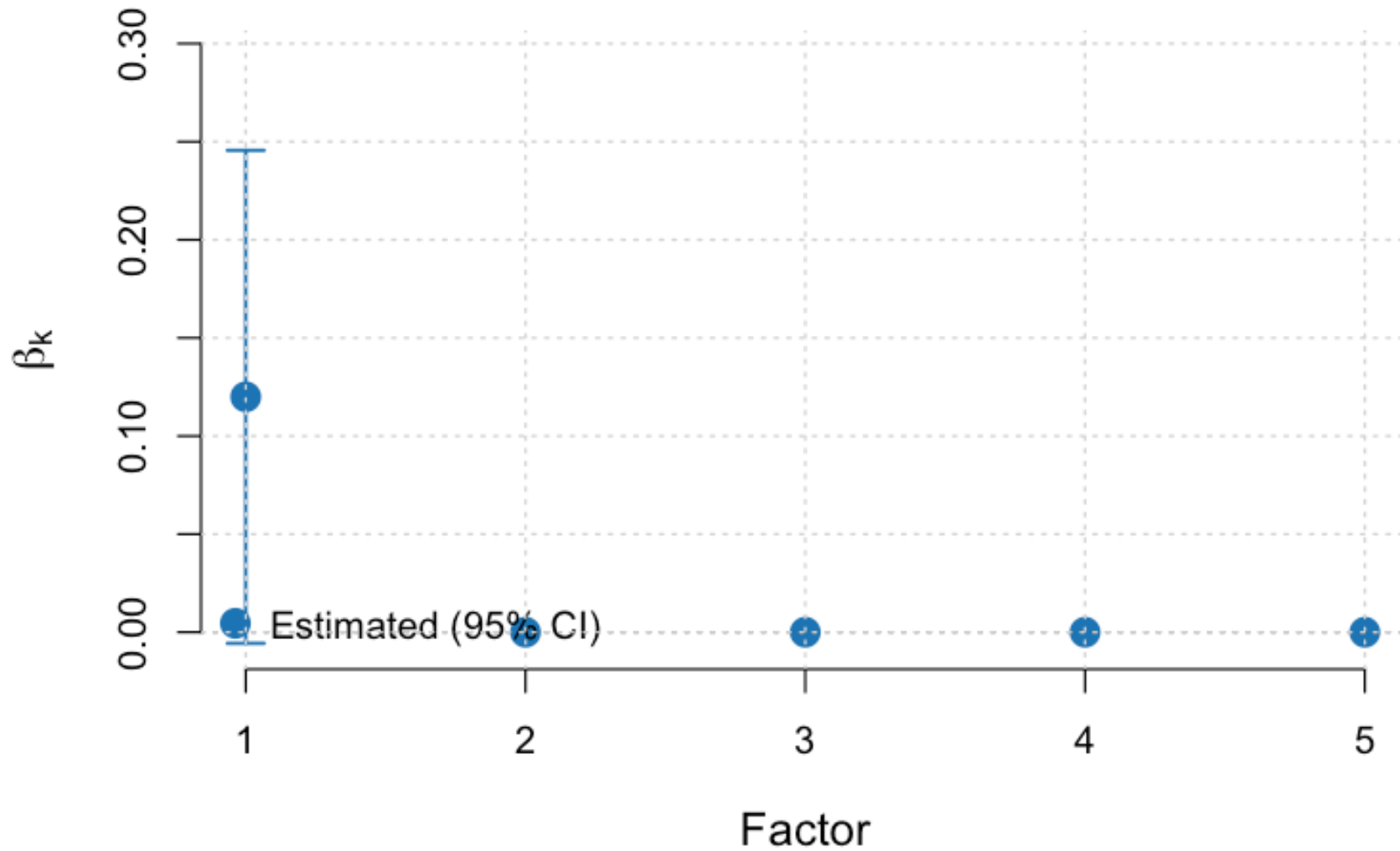


Figure 18: PACA_AU_array

**Figure 3: Estimated Survival Coefficients (95% CI)
(Moffitt_GEO_array)**

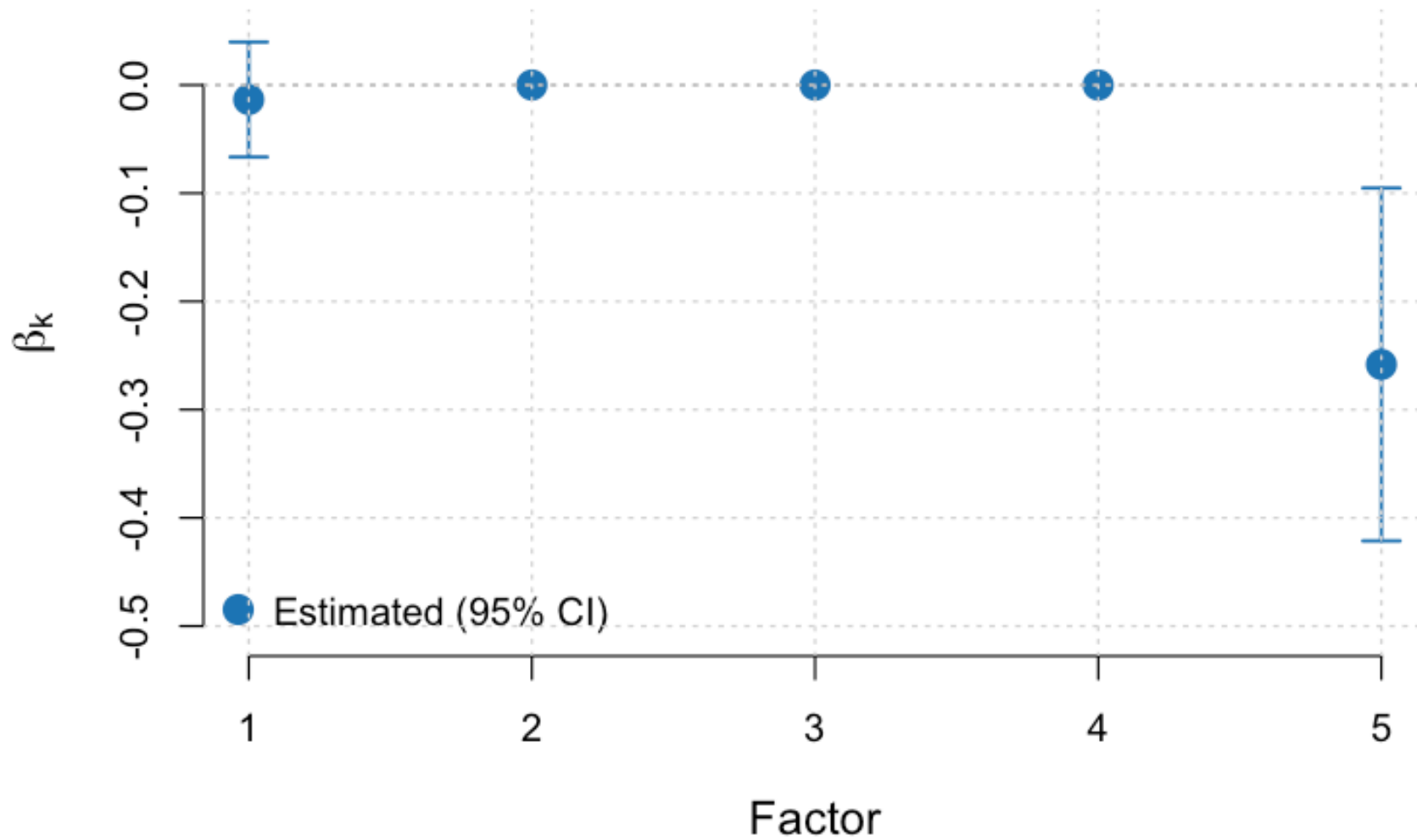


Figure 19: Moffitt

**Figure 3: Estimated Survival Coefficients (95% CI)
(PACA_AU_seq)**

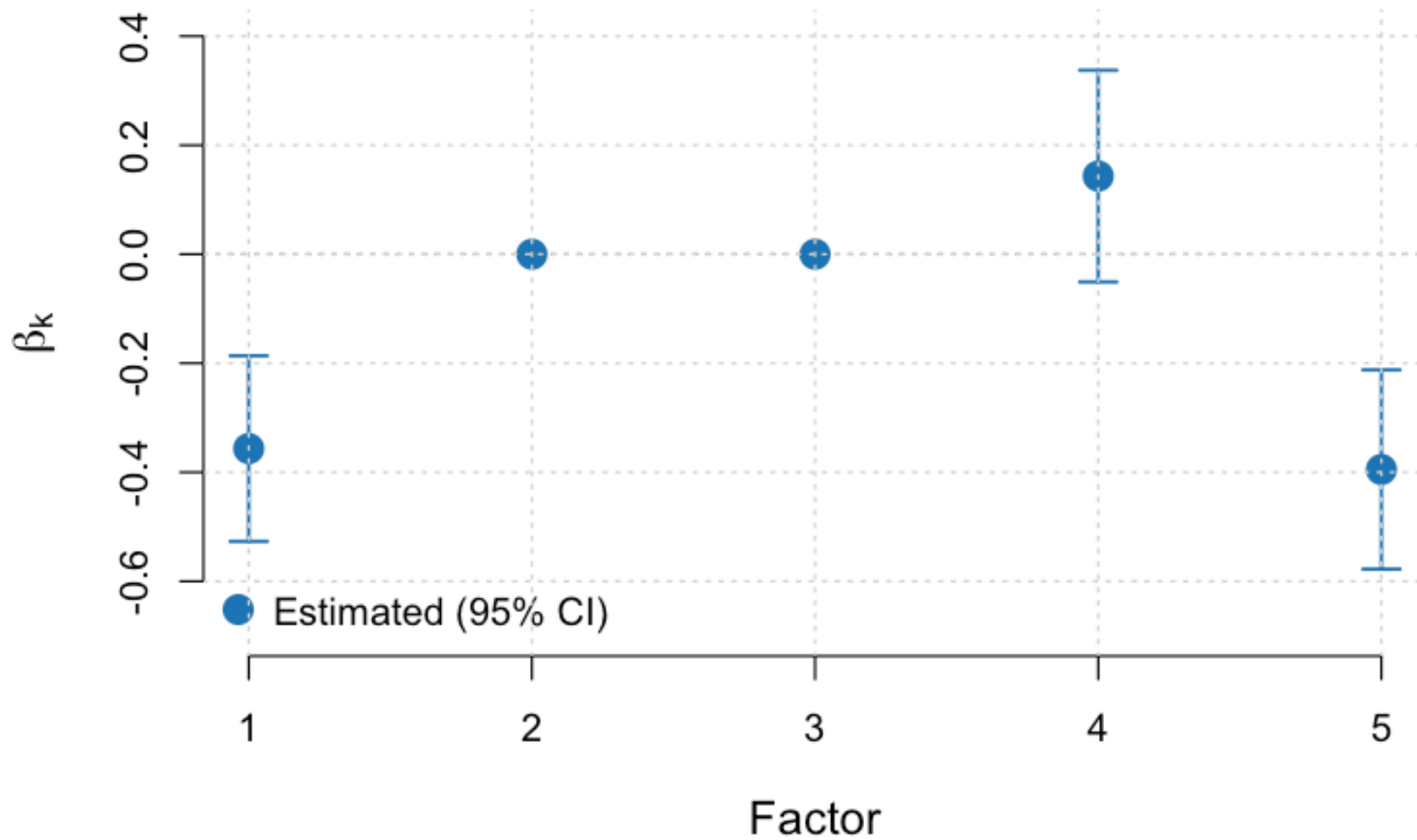


Figure 20: PACA_AU_seq

**Figure 3: Estimated Survival Coefficients (95% CI)
(CPTAC)**

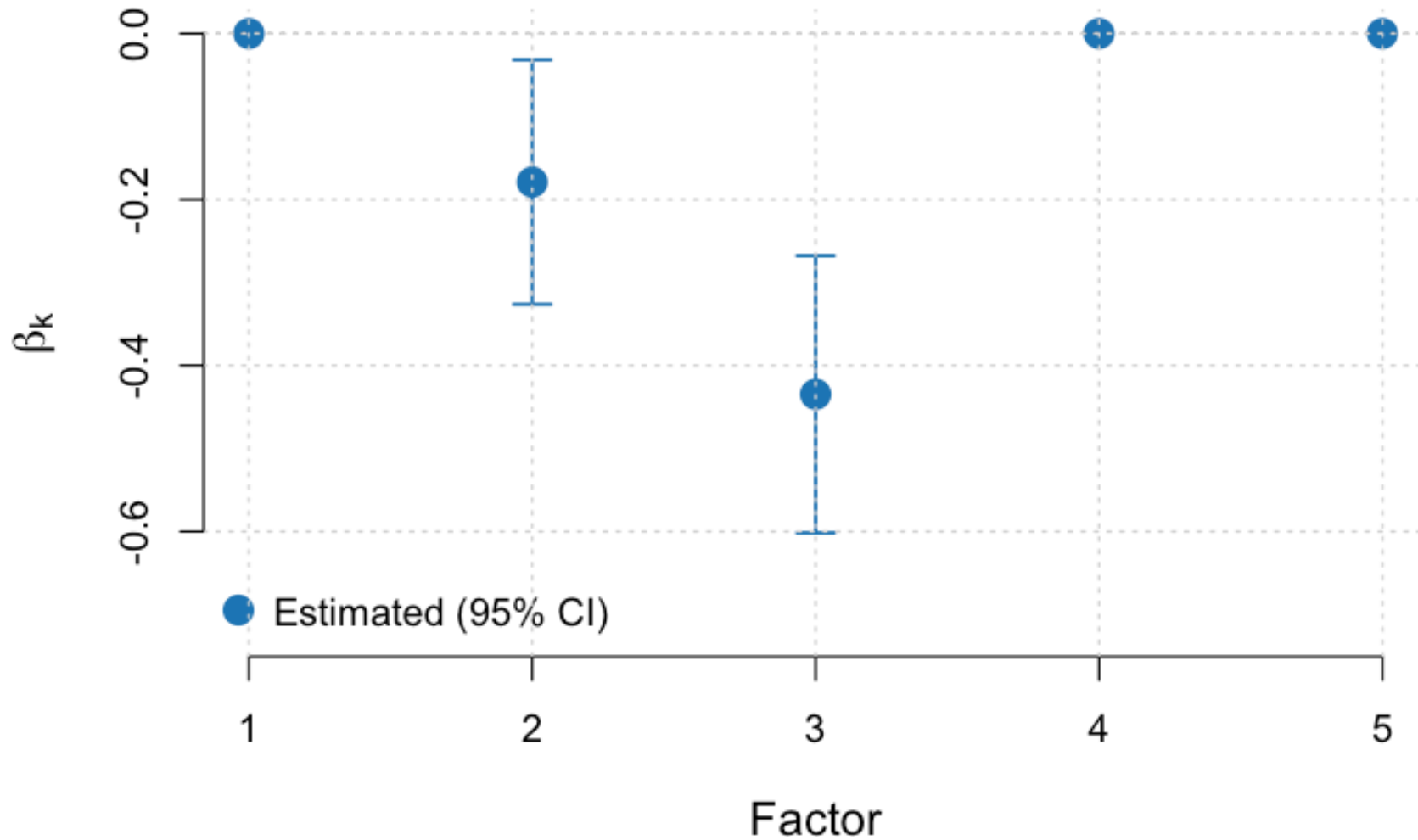


Figure 21: CPTAC

This slide is useful when someone asks whether the prognostic-factor pattern is specific to Puleo and TCGA or visible elsewhere. I would use it to show that every cohort has its own mix of null and non-null factors, but the broader separation between prognostic and expression-only structure appears repeatedly.

Per-Cohort Kaplan-Meier Curves

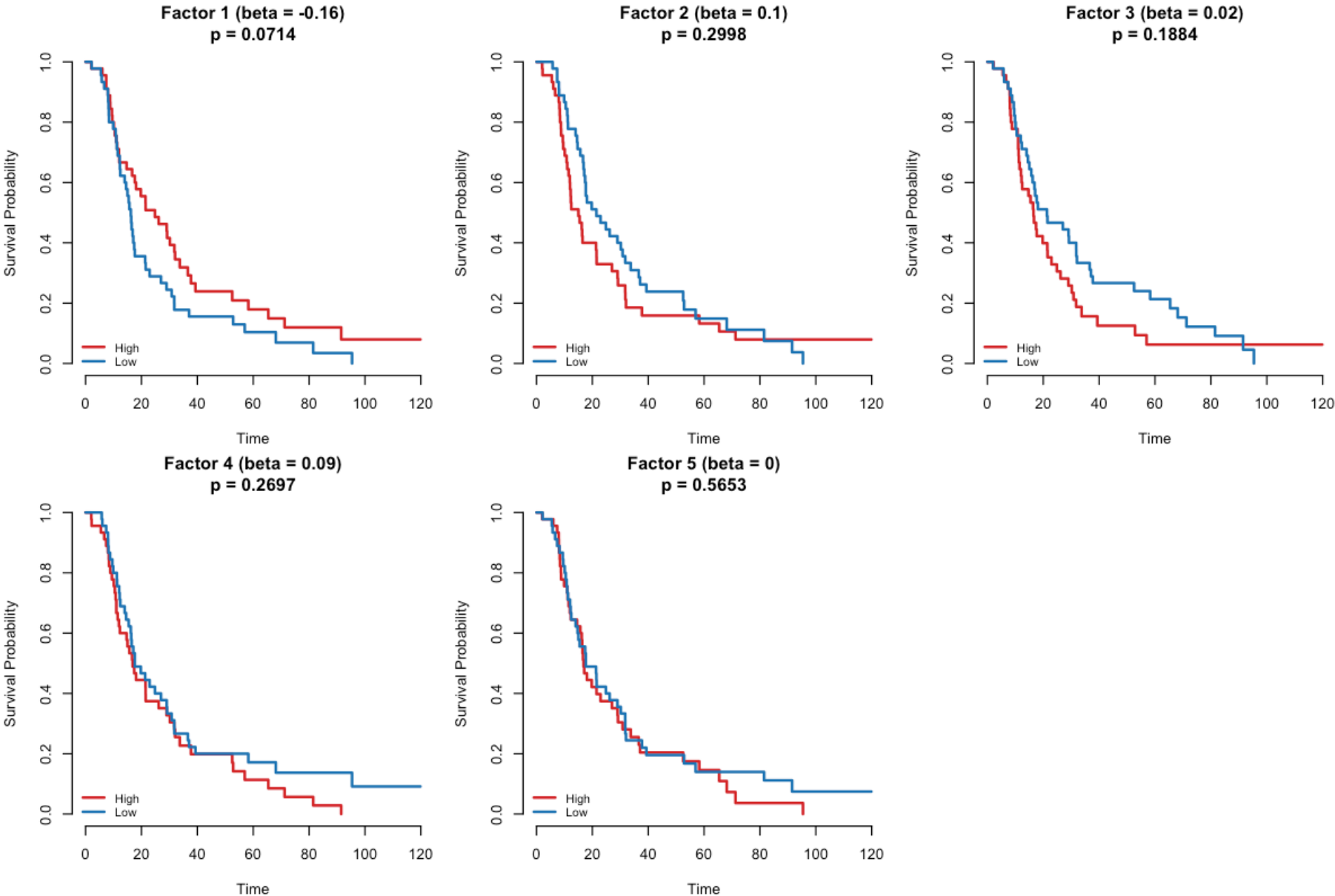


Figure 22: Dijk

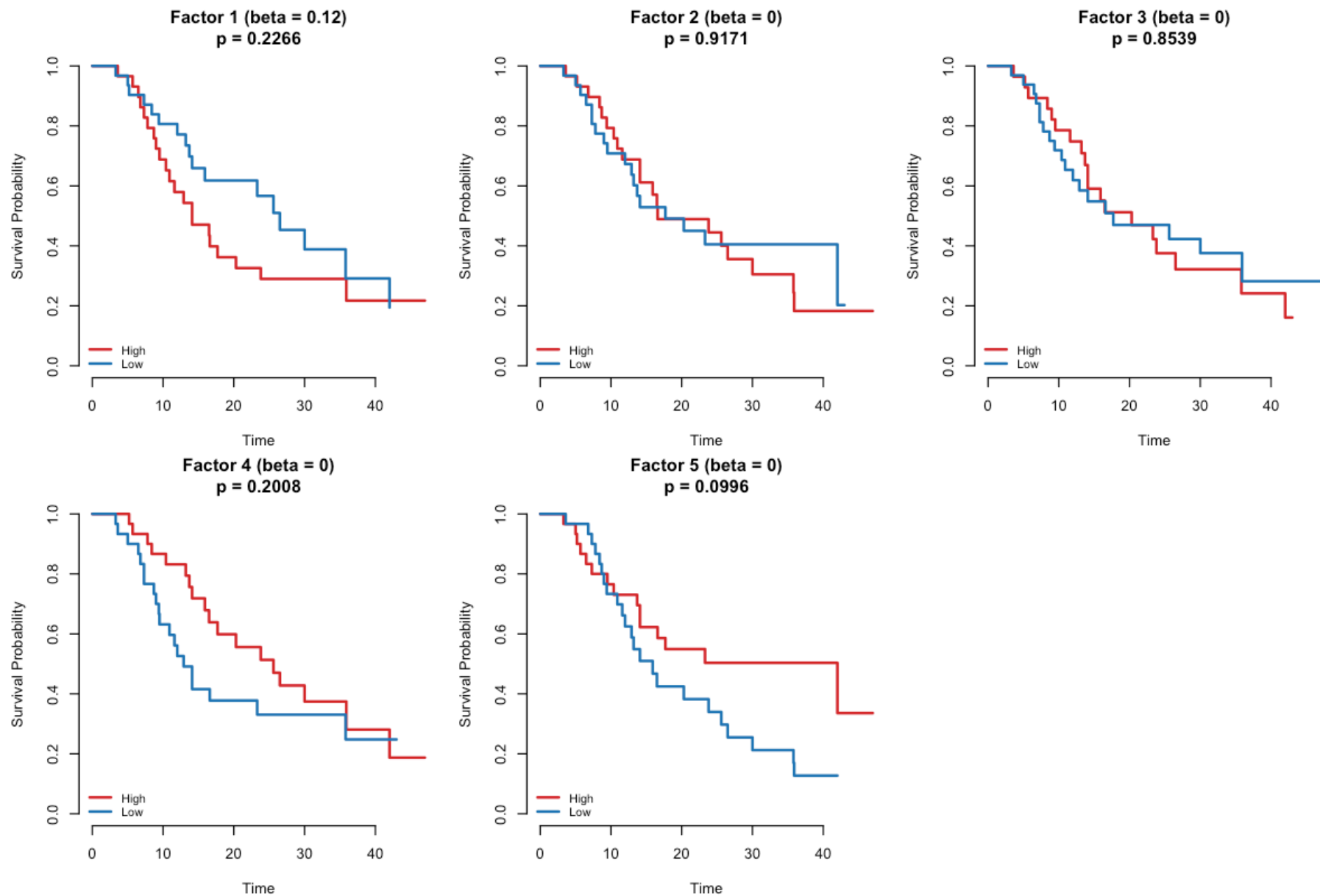


Figure 23: PACA_AU_array

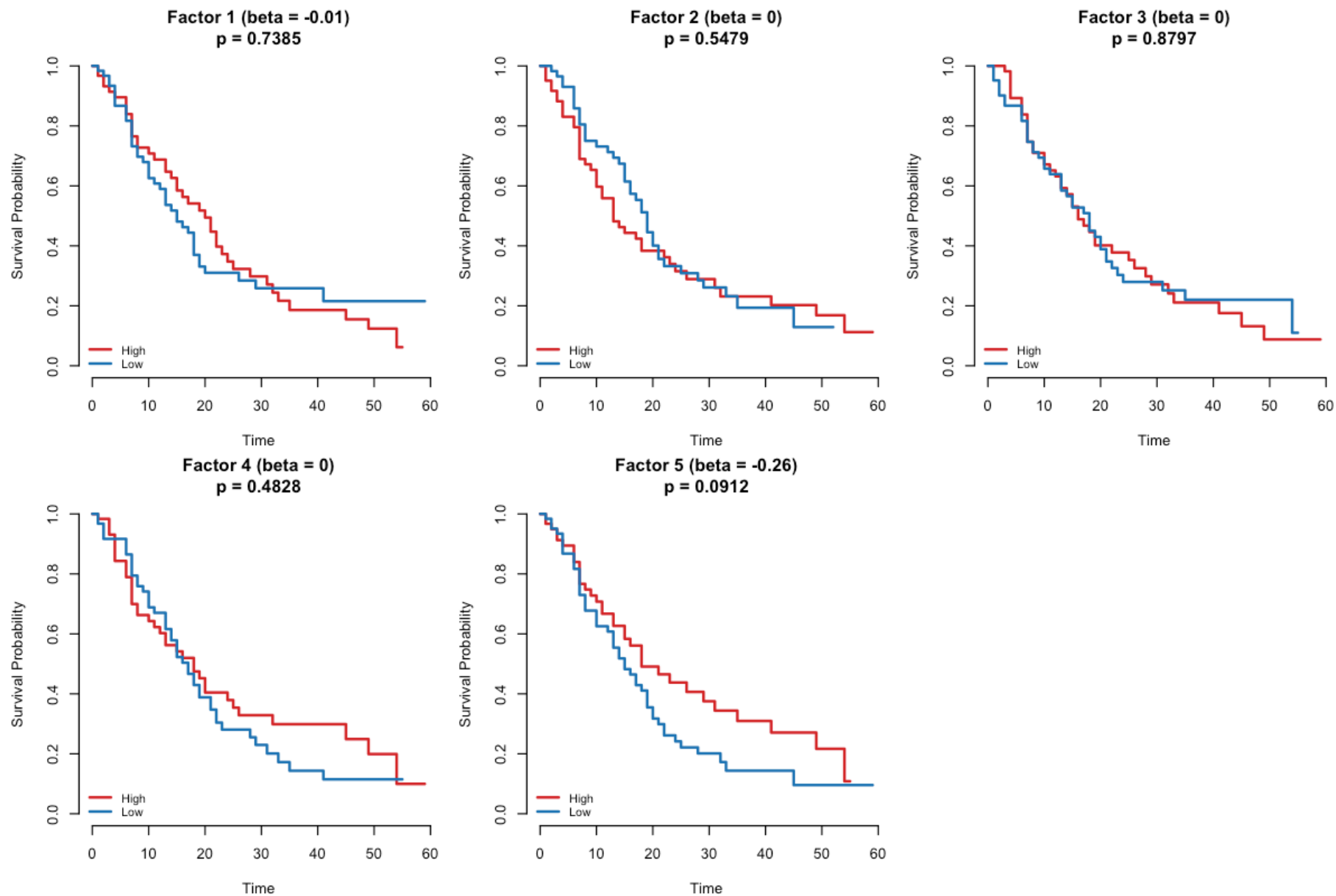


Figure 24: Moffitt

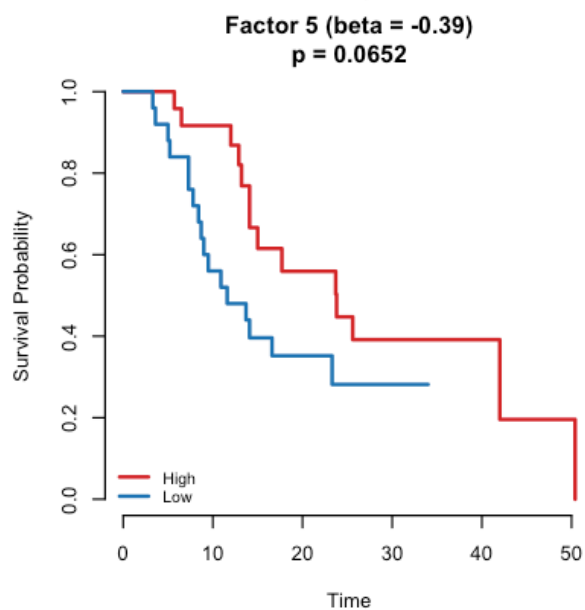
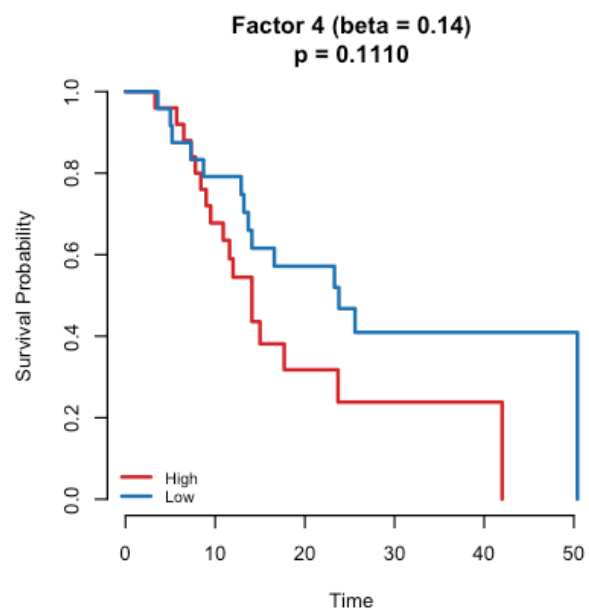
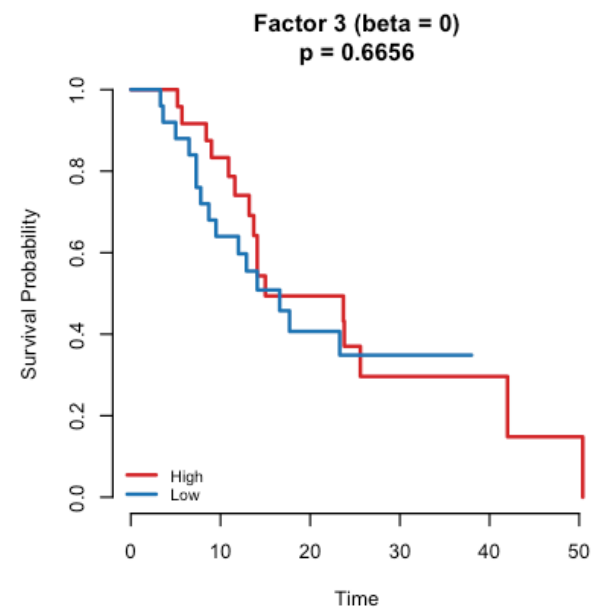
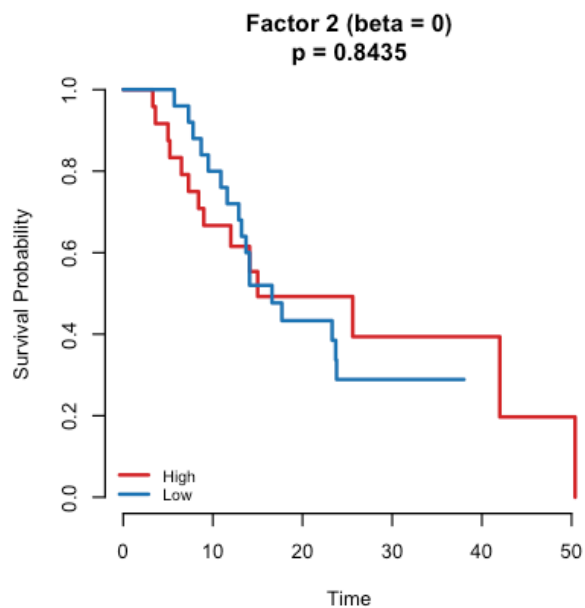
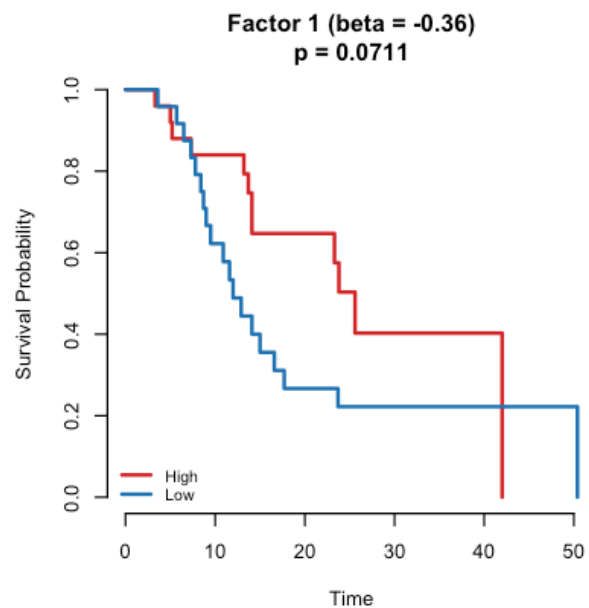


Figure 25: PACA_AU_seq

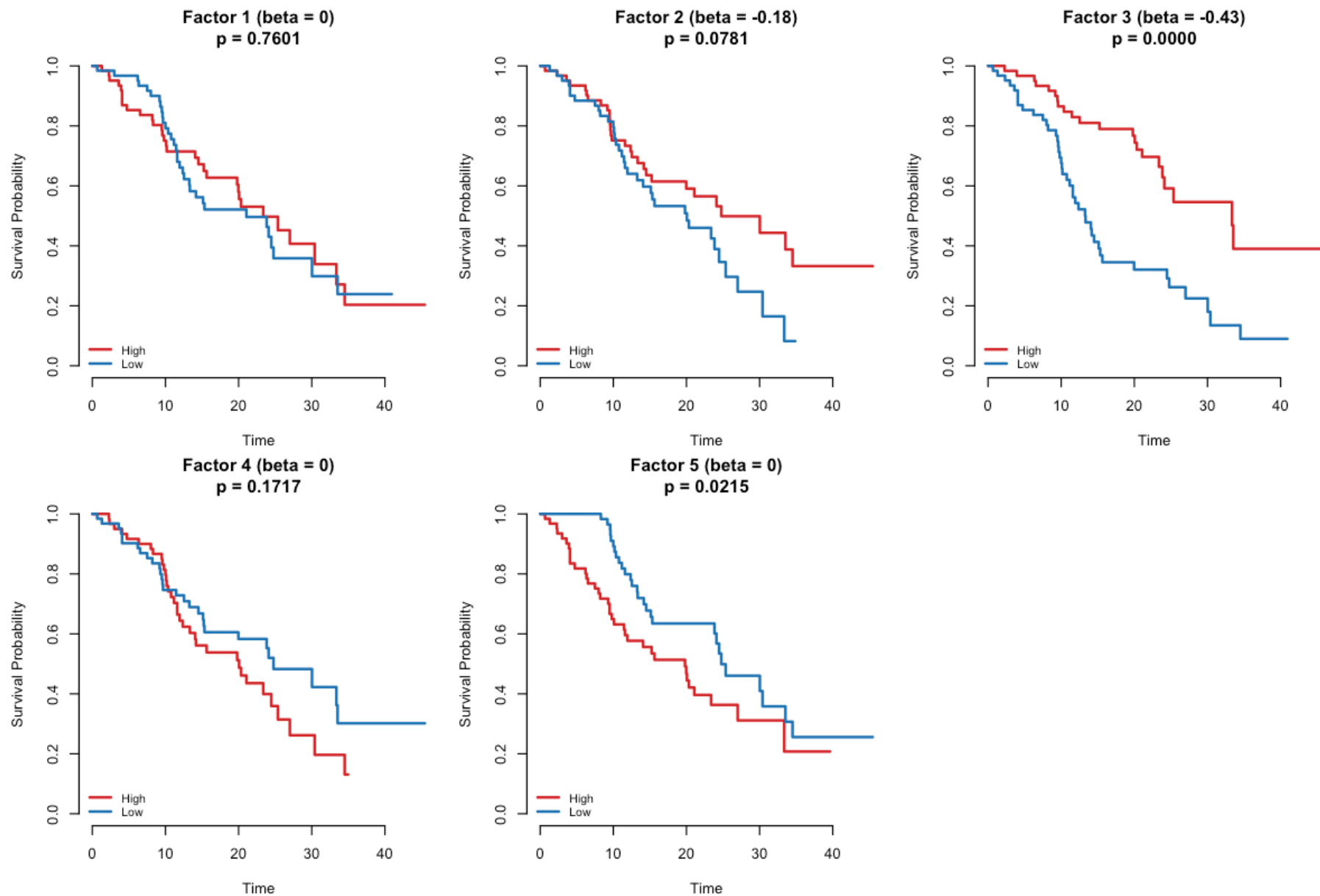
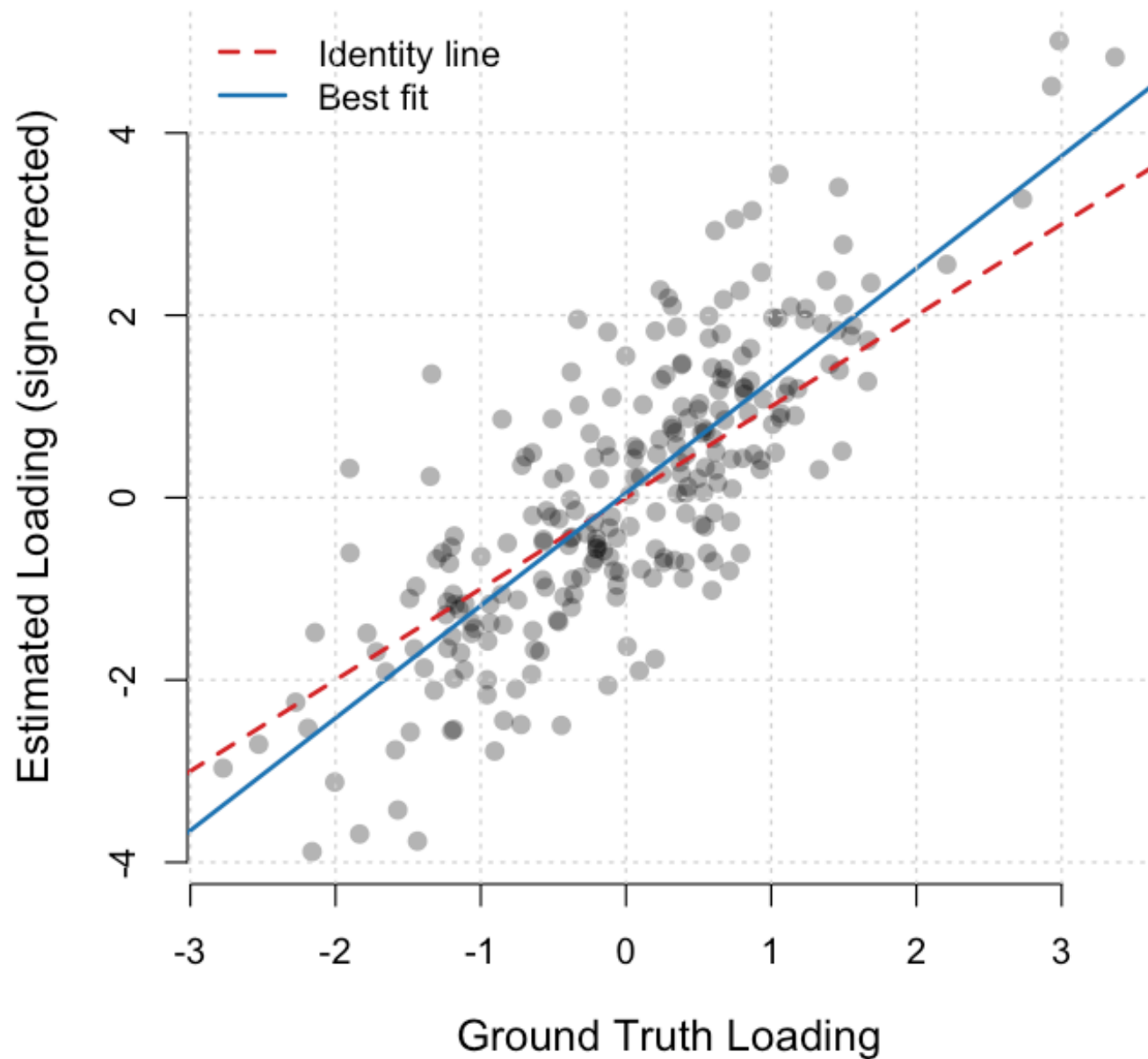


Figure 26: CPTAC

Use this when the audience wants more direct survival visualisation across the full cohort set. The point is not that every curve is equally dramatic, but that the learned programs often translate into clinically meaningful patient stratification beyond the two highlighted cohorts in the main deck.

Figure 6: Signal Recovery
(Est F3 vs True F1, $r = 0.797$)



This is the backup for anyone who wants more detail on the simulation setup or factor-recovery result. I would use it to reinforce that the method is recovering the intended latent structure, not just producing a good-looking survival metric.

Method Comparison

SBMF vs. Penalised Supervised MF Approaches

Feature	Penalised MF (e.g., deSurv)	SBMF (this work)
Inference	Penalised optimisation	Full Bayesian (CAVI)
Objective balance	Tuning parameter alpha	ELBO — natural balance
Priors	Fixed penalty lambda	EBNM — prior learns from data
Uncertainty	Point estimates only	Posterior mean + variance
K selection	CV grid search	Auto-prune (K_{eff})
Constraints	Nonnegative (NMF)	Unconstrained (allows negative loadings)
Error-in-variables	Not addressed	Uses $\text{Var}(L) + \text{E}(L)^2$ correction
Second moments	N/A	Prevents overfitting to uncertain loadings

Use this slide if the discussion turns to positioning SBFM relative to penalised supervised factorisation methods. The key message is not that one framework is universally better, but that SBFM gives us adaptive shrinkage, uncertainty-aware updates, and a more principled way to connect expression structure with survival.